Outline of Econometric Models

Methods, Tests and Estimators

Concepts, Key Equations, Real-World Applications and References

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How to Use This Outline

The chapters follow the numbering of the original outline: chapter n is section n of the outline and section $n.m$ is item $n.m$. Each item now gives

- the **idea** in plain language, before any notation;
- the **key equation, hypothesis or algorithm**;
- **A simple example**: a small numerical illustration, worked by hand, in every chapter;
- where possible a **real-world application**, either from the published literature (cited) or computed for this note on a data set that ships with R (marked R EXAMPLE; the accompanying R script, listed in the appendix, reproduces every number);
- the **references**, numbered as in the original outline. References 1–190 are the original list with their DOI or search links; references 191–205 were added for the applications and 206–259 for the new items.

Items numbered after the original ones in each chapter were added in the November 2026 revision to cover methods developed since the original outline: tests with more than two structural breaks (Chapters 4, 6 and 24), and newer tools for other topics such as local projections, proxy SVARs, bubble tests, realised volatility, staggered difference-in-differences and large Bayesian VARs. They are numbered after the original items, so the original numbering is unchanged. Each chapter closes with a short "In plain words" summary. Chapter 35 lists the corrections to the original outline, and Chapter 34 gives a map of which method to use when.

Notation. L is the lag operator ($Ly_t = y_{t-1}$), $\Delta = 1 - L$, ε_t a white-noise error, and $I(d)$ means integrated of order d (stationary after d differences).

* * *

CHAPTER 1

TRADITIONAL ECONOMETRICS

The classical linear regression model is $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, estimated by ordinary least squares (OLS), $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ [65, 185].

1.1 CLASSICAL LINEAR REGRESSION MODEL

Idea. Draw the line (or plane) through a cloud of points that makes the vertical misses as small as possible. "Linear" means linear in the *parameters*; the variables themselves may be logs, squares or products.

With n observations and K regressors (including the constant), OLS minimises $S(\beta) = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$. Setting the derivative to zero gives the *normal equations* $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$, and so the formula above; the residuals $\hat{\varepsilon} = \mathbf{y} - \mathbf{X}\hat{\beta}$ are orthogonal to every regressor, $\mathbf{X}'\hat{\varepsilon} = \mathbf{0}$. The error variance is estimated by $s^2 = \hat{\varepsilon}'\hat{\varepsilon}/(n-K)$, and $\widehat{\mathrm{Var}}(\hat{\beta}) = s^2(\mathbf{X}'\mathbf{X})^{-1}$.

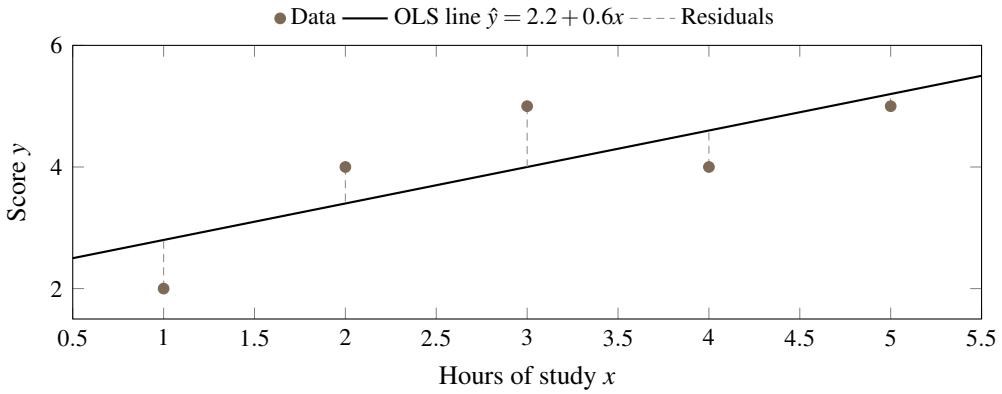


Figure 1.1: Least squares chooses the line that minimises the sum of the squared vertical distances (dashed) between the points and the line. Data from the simple example of this chapter.

1.2 ASSUMPTIONS, PROPERTIES AND GOODNESS OF FIT

The classical assumptions are: (A1) linearity in parameters; (A2) full column rank of $\mathbf{X}$ (no exact multicollinearity); (A3) strict exogeneity, $\mathrm{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$; (A4) spherical errors, $\mathrm{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2\mathbf{I}$ (homoskedastic and uncorrelated). Under A1–A4 the *Gauss–Markov theorem* says OLS is BLUE: the best (minimum-variance) linear unbiased estimator. Adding (A5) normal errors gives exact t and F tests in finite samples; without A5, the same tests hold approximately in large samples by the central limit theorem.

Goodness of fit is $R^2 = 1 - \mathrm{SSR}/\mathrm{SST}$, the share of the variation in y explained by the regressors, and the adjusted $\bar{R}^2 = 1 - (1 - R^2)(n-1)/(n-K)$, which penalises extra regressors. A high R^2 does not mean a model is right: two trending series can give $R^2 \approx 1$ with no relation at all (spurious regression, Chapter 5).

1.3 OTHER FORMS OF THE REGRESSION MODEL

- **Log–log**: $\ln y = \alpha + \beta \ln x$; β is an elasticity (a 1% rise in x changes y by β%).
- **Semi-log**: $\ln y = \alpha + \beta x$; a unit rise in x changes y by about 100β% (exactly $100(e^{\beta} - 1)$%). Wage equations use this form.
- **Polynomial and interaction**: $y = \alpha + \beta_1 x + \beta_2 x^2$ captures diminishing returns; $\beta_3 x_1 x_2$ lets the effect of x_1 depend on x_2.
- **Restricted least squares**: impose theory, e.g. constant returns to scale $\beta_K + \beta_L = 1$ in a Cobb–Douglas production function, and test it with an F test comparing restricted and unrestricted residual sums of squares.
- **Generalised and weighted least squares**: when $\mathrm{Var}(\varepsilon) = \sigma^2\Omega \neq \sigma^2\mathbf{I}$, GLS

$$\hat{\beta}_{GLS} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{y}$$

  is efficient; WLS is the special case of a diagonal Ω.

1.4 VIOLATIONS AND CORRECTIONS

Problem	Consequence	Detection and remedy
Multicollinearity	Unbiased but imprecise estimates; unstable signs	$\mathrm{VIF}_j = 1/(1-R_j^2)$ above 10; drop, combine or use ridge/PCA
Heteroskedasticity	OLS unbiased, standard errors wrong	Breusch–Pagan [23], White [184]; robust (White) s.e. or WLS
Autocorrelation	Standard errors wrong; inefficiency	Durbin–Watson [40], Breusch–Godfrey; Newey–West HAC s.e. [132] or GLS
Misspecification	Biased estimates	RESET [154]
Non-normality	Exact small-sample tests invalid	Jarque–Bera [90], $JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2_2$
Endogeneity, $\mathrm{E}[\varepsilon \mid x] \neq 0$	OLS biased and inconsistent	Instrumental variables / 2SLS; Hausman test [81]

Table 1.1: Violations of the classical assumptions.

R EXAMPLE: the Longley data. Longley's US data [191] for 1947–62, built to test the numerical accuracy of regression software, regress employment on six macroeconomic series. $R^2 = 0.9955$, yet GNP, the GNP deflator and population are all individually insignificant ($|t| < 1.1$). The variance inflation factors explain why: 1789 for GNP, 759 for the time trend, 399 for population, far above the rule-of-thumb of 10. The other diagnostics are clean: Durbin–Watson $= 2.56$, Breusch–Pagan $p = 0.87$, Jarque–Bera $p = 0.71$. The lesson: a high R^2 with insignificant t-statistics is the signature of multicollinearity.

In the original paper. *Durbin and Watson* [40] showed that the exact distribution of $d = \sum(\hat\varepsilon_t - \hat\varepsilon_{t-1})^2 / \sum \hat\varepsilon_t^2 \approx 2(1-\hat\rho)$ depends on the regressors, so they derived lower and upper bounds d_L, d_U valid for any $\mathbf{X}$, leaving an inconclusive zone; the test is not valid when a lagged dependent variable is a regressor. *Breusch and Pagan* [23] derived a Lagrange multiplier test for $\sigma_t^2 = h(\mathbf{z}_t'\alpha)$: half the explained sum of squares of a regression of $\hat\varepsilon_t^2/\hat\sigma^2$ on $\mathbf{z}_t$, asymptotically χ^2, and, importantly, the same whatever the form of h. *White* [184] showed that $(\mathbf{X}'\mathbf{X})^{-1}(\sum \hat\varepsilon_t^2 \mathbf{x}_t\mathbf{x}_t')(\mathbf{X}'\mathbf{X})^{-1}$ estimates the covariance matrix consistently under heteroskedasticity of unknown form, and built a direct test by comparing it with the usual estimator, computed as nR^2 from a regression of $\hat\varepsilon_t^2$ on the regressors, their squares and cross-products. *Jarque and Bera* [90] obtained their statistic as an LM test of normality within the Pearson family of distributions, so it is efficient against skewed and fat-tailed departures.

1.5 DUMMY VARIABLES AND THEIR TYPES

A dummy is a 0/1 variable for a qualitative factor.

- **Intercept dummy**: $y = \alpha + \delta D + \beta x$ shifts the line.
- **Slope dummy**: $y = \alpha + \beta x + \gamma (D \times x)$ changes its slope.
- **Seasonal dummies**: 11 monthly or 3 quarterly dummies with a constant.

With k categories use $k-1$ dummies; including all k with a constant makes the columns exactly collinear (the *dummy-variable trap*). The *Chow test* [32] checks whether all coefficients differ between two subsamples split at a known date T_B:

$$F = \frac{(\mathrm{SSR}_P - \mathrm{SSR}_1 - \mathrm{SSR}_2)/K}{(\mathrm{SSR}_1 + \mathrm{SSR}_2)/(n-2K)} \sim F(K, n-2K), \tag{1.1}$$

equivalent to adding an intercept dummy and slope dummies for every regressor.

R EXAMPLE: the UK seat-belt law. Wearing front seat belts became compulsory in Great Britain on 31 January 1983. Regressing log monthly car-driver deaths (1969–84) on a law dummy, log petrol price, log distance driven, a trend and 11 monthly dummies gives a law coefficient of -0.133 (s.e. 0.034): deaths fell by $100(e^{-0.133}-1) = -12.4\%$, other things equal. This data set comes from the classic structural time-series study of the law by Harvey and Durbin [192].

In the original paper. Chow [32] proposed two tests. The first is the F test above, comparing one regression on the pooled sample with separate regressions on the two sub-samples. The second, the *predictive* test, is for the case where the second sample is too short to estimate the model ($n_2 < K$): fit the model on the first sample, predict the second, and test whether the prediction errors are larger than sampling variation allows, $F = [(\mathrm{SSR}_P - \mathrm{SSR}_1)/n_2]/[\mathrm{SSR}_1/(n_1 - K)]$.

1.6 SIMULTANEOUS EQUATION MODELS AND IDENTIFICATION

Idea. Price and quantity are set together by supply *and* demand, so a regression of quantity on price traces neither curve. This is the identification problem first posed for demand curves by Working [193] and formalised by Haavelmo [69].

The structural form $\Gamma \mathbf{y}_t + \mathbf{B}\mathbf{x}_t = \mathbf{u}_t$ has the reduced form $\mathbf{y}_t = \Pi \mathbf{x}_t + \mathbf{v}_t$ with $\Pi = -\Gamma^{-1}\mathbf{B}$; the reduced form can always be estimated by OLS, the question is whether the structural parameters can be recovered from Π. For an equation in a system of G equations:

- **Order condition** (necessary): the number of exogenous variables excluded from the equation must be at least the number of included endogenous variables minus one.
- **Rank condition** (necessary and sufficient): the matrix of coefficients of the excluded variables in the other equations has rank $G-1$.

Estimators: two-stage least squares (2SLS: regress each endogenous regressor on all exogenous variables, then use the fitted values), three-stage least squares (3SLS: 2SLS plus GLS across equations) and limited-information maximum likelihood (LIML) [65].

Application. In a market for an agricultural good, rainfall shifts supply but not demand. It is excluded from the demand equation, so it identifies the demand curve: rainfall moves the supply curve along a stable demand curve and traces it out.

1.7 CLUSTER-ROBUST INFERENCE

When errors are correlated within groups (employees in the same firm, loans in the same branch, repeated observations of one borrower), ordinary and even White standard errors are too small. The cluster-robust variance $\hat{\mathbf{V}} = (\mathbf{X}'\mathbf{X})^{-1}\big(\sum_g \mathbf{X}_g'\hat{\boldsymbol{\varepsilon}}_g\hat{\boldsymbol{\varepsilon}}_g'\mathbf{X}_g\big)(\mathbf{X}'\mathbf{X})^{-1}$ allows arbitrary correlation within each cluster g. Cameron, Gelbach and Miller extend it to two-way clustering (e.g. by firm and by month) [256]. With few clusters (under about 30–50), use the wild cluster bootstrap.

1.8 WEAK INSTRUMENTS

IV is only as good as its first stage. When instruments are weakly correlated with the endogenous regressor, 2SLS is biased towards OLS and its t-tests are badly sized even in large samples [257]. The rule of thumb "first-stage $F > 10$" comes from this literature; with heteroskedastic or clustered errors the effective F statistic of Montiel Olea and Pflueger should be used instead [258].

A SIMPLE EXAMPLE. Five students report hours of study x and test scores y (out of 10): $x = (1,2,3,4,5)$, $y = (2,4,5,4,5)$. The means are $\bar{x} = 3$, $\bar{y} = 4$.

$$\hat{\beta} = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2} = \frac{4+0+0+0+2}{4+1+0+1+4} = \frac{6}{10} = 0.6, \qquad \hat{\alpha} = 4 - 0.6 \times 3 = 2.2.$$

Each extra hour adds 0.6 marks. Fitted scores are $2.8, 3.4, 4.0, 4.6, 5.2$; residuals $-0.8, 0.6, 1.0, -0.6, -0.2$ (they sum to zero); SSR $= 2.4$, SST $= 6$, so $R^2 = 1 - 2.4/6 = 0.6$: study hours explain 60% of the variation in scores.

> IN PLAIN WORDS. Regression draws the best line through the data; the assumptions say when that line can be trusted. When they fail, the line still exists but its standard errors, or even its slope, can mislead.

* * *

CHAPTER 2

TIME SERIES ECONOMETRICS

A series is *weakly stationary* if its mean, variance and autocovariances $\gamma_k = \mathrm{Cov}(y_t, y_{t-k})$ do not depend on t. By Wold's decomposition, any purely non-deterministic stationary process has an infinite moving-average representation $y_t = \mu + \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ [72]: every stationary series is, at heart, a weighted sum of past shocks.

2.1 UNIVARIATE TIME SERIES

A series is decomposed into *trend* (long-run movement), *seasonal* (calendar pattern) and *irregular* components. The autocorrelation function (ACF) $\rho_k = \gamma_k / \gamma_0$ measures total correlation with lag k; the partial autocorrelation function (PACF) measures the correlation with lag k after removing the effect of lags $1, \ldots, k-1$. Together they identify ARMA orders [21]:

Model	ACF	PACF
AR(p)	decays gradually	cuts off after lag p
MA(q)	cuts off after lag q	decays gradually
ARMA(p,q)	decays after lag q	decays after lag p

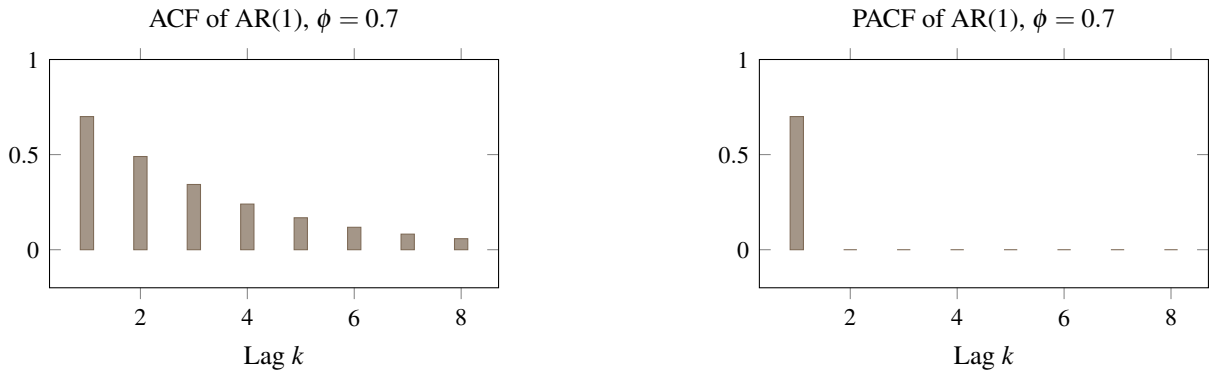


Figure 2.1: The fingerprints of an AR(1): the ACF decays geometrically (0.7^k), the PACF has a single spike at lag 1. An MA(1) shows the mirror image.

2.2 AUTOREGRESSIVE MODEL, AR(p)

$\phi(L)y_t = c + \varepsilon_t$ with $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$. Today is a weighted sum of the past plus a shock. The process is stationary if and only if all roots of $\phi(z) = 0$ lie outside the unit circle; for AR(1), $|\phi_1| < 1$. Its mean is $c/(1 - \sum \phi_i)$, and a shock decays at rate ϕ_1^k in the AR(1): the *half-life* is $\ln 0.5 / \ln \phi_1$ periods.

Example. Deviations of real exchange rates from purchasing power parity are often modelled as AR(1) with annual $\phi_1 \approx 0.85$, a half-life of about four years, the well-known "PPP puzzle" [198].

2.3 MOVING AVERAGE MODEL, MA(q)

$y_t = \mu + \theta(L)\varepsilon_t$, $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$. A shock affects the series for exactly q periods and then vanishes. A finite MA is always stationary; it is *invertible* (can be written as an AR(∞), so shocks can be recovered from the data) if the roots of $\theta(z) = 0$ lie outside the unit circle. Overlapping observations create MA errors: monthly data on 3-month returns follow an MA(2).

2.4 ARMA(p,q)

$\phi(L)y_t = c + \theta(L)\varepsilon_t$. Combining both parts is *parsimonious*: an ARMA(1,1) can mimic an AR or MA of high order with just two parameters.

2.5 ARIMA(p,d,q)

$\phi(L)(1-L)^d y_t = c + \theta(L)\varepsilon_t$: difference the series d times to remove d unit roots, then fit an ARMA. Seasonal ARIMA adds factors in L^s.

2.6 BOX–JENKINS METHODOLOGY

Four steps, repeated until the model is adequate [21]:

1. **Identification**: transform (logs) and difference to stationarity; read the ACF/PACF; compare candidates by AIC $= -2\ln L + 2k$ [2] or BIC $= -2\ln L + k\ln n$ [160].
2. **Estimation** by maximum likelihood.
3. **Diagnostic checking**: residuals should be white noise, checked by the Ljung–Box statistic $Q(m) = n(n+2)\sum_{k=1}^{m} \hat{r}_k^2/(n-k) \sim \chi^2_{m-p-q}$ [112].
4. **Forecasting** from the fitted model, with widening intervals.

R EXAMPLE: the airline model. Box and Jenkins' own example is monthly international airline passengers, 1949–60. For the logged series they proposed $(1-L)(1-L^{12})\ln y_t = (1+\theta L)(1+\Theta L^{12})\varepsilon_t$, now called the *airline model*. Re-estimating it gives $\hat{\theta} = -0.40$ (s.e. 0.09) and $\hat{\Theta} = -0.56$ (s.e. 0.07); the Ljung–Box test on 24 residual autocorrelations has $p = 0.23$, so the residuals pass as white noise. The model forecasts 450 thousand passengers for January 1961 rising to a July peak of 670 thousand.

2.7 AUTOMATIC MODEL SELECTION

Box–Jenkins identification by eye does not scale to thousands of series (every product in a retailer, every branch of a bank). Hyndman and Khandakar's algorithm automates it: choose the differencing order by unit-root (KPSS) tests, then search ARIMA orders stepwise by AIC [259]. It is the engine of `auto.arima` in R. Its output should still go through the diagnostic step.

A SIMPLE EXAMPLE. Let $y_t = 2 + 0.5y_{t-1} + \varepsilon_t$. The long-run mean is $2/(1-0.5) = 4$. If this year $y = 10$, the forecasts for the next four years are $2 + 0.5 \times 10 = 7$, then 5.5, 4.75, 4.375: the gap to the mean $(6, 3, 1.5, 0.75, 0.375)$ halves every year, a half-life of exactly one period $(\ln 0.5/\ln 0.5 = 1)$. With $\phi = 0.9$ instead, the half-life would be $\ln 0.5/\ln 0.9 = 6.6$ years.

IN PLAIN WORDS. Box–Jenkins modelling is detective work: look at the correlation fin-

gerprints, propose a small model, check that what is left over is pure noise, then forecast.

* * *

CHAPTER 3

UNIT ROOT TESTS WITHOUT BREAKS

If $y_t = \rho y_{t-1} + \varepsilon_t$ with $\rho = 1$ (a random walk), shocks are permanent and the variance grows without bound. OLS t-statistics then follow non-standard distributions (functionals of Brownian motion), so special critical values are needed. Why it matters: regressing one random walk on another independent one gives "significant" results most of the time (spurious regression), and forecasts of a unit-root series never revert to a mean.

The landmark study. Nelson and Plosser [194] tested 14 long annual US macroeconomic series and could not reject a unit root in 13 of them, overturning the view that output simply fluctuates around a deterministic trend. Much of Chapters 3 and 4 is the response to that paper.

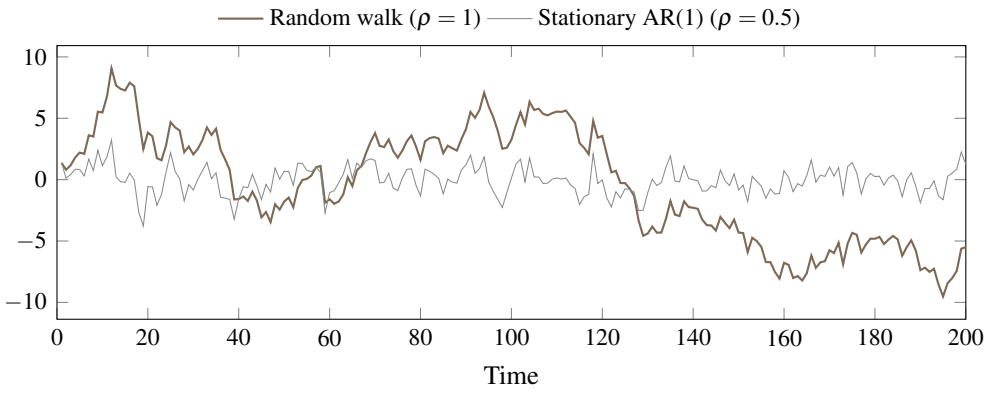


Figure 3.1: The same 200 shocks fed into a random walk and into a stationary AR(1). The random walk wanders away and never returns; the AR(1) keeps crossing its mean of zero.

3.1 DICKEY–FULLER TEST

Subtract y_{t-1} from both sides: $\Delta y_t = \alpha + \gamma t + \delta y_{t-1} + \varepsilon_t$ with $\delta = \rho - 1$, and test H_0: $\delta = 0$ (unit root) against H_1: $\delta < 0$ (stationary) [37, 52]. The statistic is the usual t-ratio, but compared with Dickey–Fuller critical values, which depend on the deterministic terms: at 5% in large

samples about -1.95 (none), -2.86 (constant) and -3.41 (constant and trend). Choosing the deterministics matters: include a trend if the series trends under the alternative.

In the original paper. Dickey and Fuller [37] studied $y_t = \rho y_{t-1} + e_t$ with three versions: no constant, a constant, and a constant and trend. They showed that when $\rho = 1$ the OLS estimator is *super-consistent*: $T(\hat{\rho} - 1)$, not $\sqrt{T}(\hat{\rho} - 1)$, has a limiting distribution, and it is skewed to the left. The "t-statistic" $\hat{\tau}$ does not follow a t or normal distribution in any of the three models. Its distribution has no closed form, so they tabulated percentiles by Monte Carlo (the tables in Fuller [52]). A key property: in the models with a constant (and trend), the null distribution does not depend on the values of the constant or trend, so the tables can be used without knowing them.

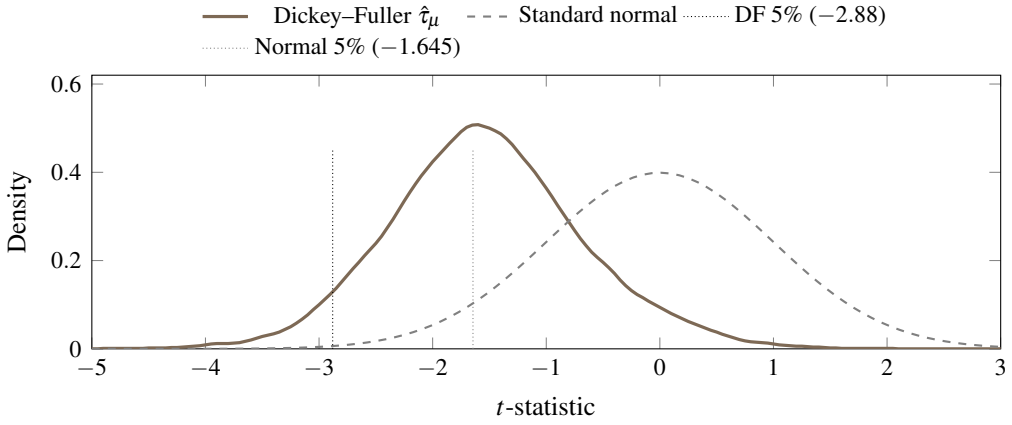


Figure 3.2: Distribution of the Dickey–Fuller t-statistic (constant included, $T = 100$, 20,000 simulated random walks) against the standard normal. It is shifted to the left, so its 5% critical value is -2.88, not -1.645; using normal tables would find "stationarity" far too often.

3.2 AUGMENTED DICKEY–FULLER TEST

Adds lagged differences, $\Delta y_t = \alpha + \gamma t + \delta y_{t-1} + \sum_{j=1}^{p} \psi_j \Delta y_{t-j} + \varepsilon_t$, so that the errors are white noise; the critical values are unchanged [37]. Lag length by AIC/BIC, or general-to-specific (start large, drop insignificant last lags).

In the original paper. Said and Dickey [264] showed that when the errors follow an ARMA process of unknown order, an autoregression in differences with k lags approximates it well enough, provided k grows with the sample, but slowly ($k^3/T \to 0$). The ADF $\hat{\tau}$ then has the same limiting distribution as the Dickey–Fuller $\hat{\tau}$, so the original tables remain valid. This is the theoretical justification for adding lags rather than modelling the error process exactly.

3.3 PHILLIPS–PERRON TEST

Instead of adding lags, PP runs the simple DF regression and corrects the statistic non-parametrically for serial correlation and heteroskedasticity using a long-run variance estimate (Newey–West type) [151, 152]. Robust to general error structures, but has poor size when the errors have a large negative MA component.

In the original paper. Phillips [151] derived the asymptotics of unit-root regressions with a functional central limit theorem, allowing weakly dependent and heterogeneously distributed errors. He showed that the Dickey–Fuller statistics then depend on two nuisance parameters: the short-run variance $\sigma^2 = \lim T^{-1} \sum \mathrm{E}(e_t^2)$ and the long-run variance $\lambda^2 = \lim T^{-1} \mathrm{E}(\sum e_t)^2$. They coincide only for white noise. The $Z(\alpha)$ and $Z(t)$ statistics subtract a correction proportional to $(\hat{\lambda}^2 - \hat{\sigma}^2)$, with $\hat{\lambda}^2$ a Newey–West-type estimate, so the Dickey–Fuller tables apply again. Phillips and Perron [152] extended this to models with drift and trend, and their simulations showed the size problems when the errors have a large negative moving-average root.

3.4 KPSS TEST

Reverses the null. Write $y_t = \xi t + r_t + \varepsilon_t$ with a random walk $r_t = r_{t-1} + u_t$; H_0: $\sigma_u^2 = 0$ (trend-stationary). The LM statistic is $\eta = T^{-2} \sum_t S_t^2 / \hat{\sigma}_{LR}^2$, where S_t are partial sums of residuals from a regression on a constant (and trend) [104]. Large partial sums (the residuals wander) reject stationarity. Used with ADF for *confirmatory analysis*: ADF rejects and KPSS does not $\Rightarrow$ stationary; ADF does not and KPSS rejects $\Rightarrow$ unit root; both or neither $\Rightarrow$ the data are not informative. Applied to the Nelson–Plosser series, KPSS found that for several of them trend-stationarity could not be rejected either.

In the original paper. Kwiatkowski, Phillips, Schmidt and Shin [104] started from a concern: standard tests take the unit root as the null and, with low power, “fail to reject” it too readily, so the evidence for unit roots in macro data may reflect only low power. Their components model writes the series as a deterministic trend, a random walk and a stationary error, so stationarity is the hypothesis that the random walk has zero variance. The statistic is the LM (locally best invariant) test of that hypothesis under Gaussian errors, corrected for serial correlation with a Bartlett-window estimate of the long-run variance. Applying it to the Nelson–Plosser series, they found that trend-stationarity could not be rejected for many of them, and concluded that the data are often not informative enough to settle the question.

3.5 NG–PERRON TESTS

The original outline repeats ARIMA here; the usual fifth member of this group is the Ng–Perron family MZ_α, MZ_t, MSB, MP_T. These modified statistics combine GLS detrending with a modified AIC (MAIC) for the lag length and have much better size than ADF and PP when the errors have a negative moving-average root [133].

In the original paper. Ng and Perron [133] diagnosed why unit-root tests over-reject when the errors have a large negative moving-average root: AIC and BIC choose too few lags. Their modified AIC (MAIC) makes the penalty depend on how far the estimated autoregression is from a unit root, so more lags are chosen precisely in the problematic case. Combined with GLS detrending and the modified Phillips–Perron statistics, the resulting tests have close to correct size and power near the efficiency envelope.

3.6 DF-GLS TEST

Elliott, Rothenberg and Stock [41] first estimate the deterministic terms by GLS under a local alternative $\bar\rho = 1 + \bar c/T$ ($\bar c = -7$ with a constant, -13.5 with a trend), subtract them, then run ADF on the detrended series without deterministics. The test is nearly efficient: its power is close to the best possible against alternatives near the unit root.

In the original paper. Elliott, Rothenberg and Stock [41] derived the asymptotic *power envelope*: the best power any test can achieve against each local alternative $\rho = 1 + c/T$. No test attains it everywhere (there is no uniformly most powerful unit-root test), but the point-optimal test that is best at $\bar c = -7$ (constant) or $\bar c = -13.5$ (trend), the alternatives at which the envelope has 50% power, comes very close to it over the whole range. DF-GLS is a feasible version: GLS detrending at $\bar c$, then a Dickey–Fuller regression without deterministic terms.

3.7 UNION-OF-REJECTIONS TESTING

Unit-root tests are sensitive to whether a trend is included and to the size of the initial observation. Harvey, Leybourne and Taylor reject the unit root if *either* a test with a constant or a test with a trend rejects, with critical values scaled up to keep the overall size at 5% [224]. This is a safe default when the deterministic specification is uncertain.

3.8 EXPLOSIVE ROOTS AND BUBBLES

Reverse the alternative: test H_0: $\rho = 1$ against H_1: $\rho > 1$ (explosive). Phillips, Shi and Yu's generalised sup ADF (GSADF) test runs right-tailed ADF regressions on expanding *and* rolling windows and takes the supremum, so it can detect several bubbles in one sample and date when each began and collapsed [223]. Applied to the S&P 500 price–dividend ratio since 1871, it dates episodes including the 1990s dot-com bubble. Central banks and regulators use it to monitor housing and asset prices.

A SIMPLE EXAMPLE. Hit two series with a shock of 10. A stationary AR(1) with $\phi = 0.5$ responds $10, 5, 2.5, 1.25, \ldots$ and returns to normal; a random walk ($\phi = 1$) responds $10, 10, 10, \ldots$ for ever. Now a Dickey–Fuller regression with a constant gives $\hat{\rho} = 0.95$, so $\hat{\delta} = -0.05$ with standard error 0.02: $t = -0.05/0.02 = -2.5$. A normal table would call this significant, but the Dickey–Fuller 5% critical value is -2.86, so the unit root *cannot* be rejected.

> IN PLAIN WORDS. A unit root means shocks never wear off. Testing for one is hard because a root of 0.98 and a root of 1 look almost the same in a finite sample; the different tests trade size against power in different ways, so use more than one.

* * *

CHAPTER 4

UNIT ROOT TESTS WITH BREAKS

A one-off shift in the level or slope of a stationary series makes ADF think shocks are permanent: the test confuses "one big permanent change" with "every shock is permanent". The tests below allow for the break.

4.1 PERRON (1989)

Perron [143] treated the 1929 crash (a level shift) and the 1973 oil price shock (a change in slope) as known, *exogenous* break dates T_B. He augmented ADF with a level-shift dummy $DU_t = 1(t > T_B)$, a trend-shift dummy $DT_t = (t - T_B)1(t > T_B)$ and a one-time pulse $D(T_B)_t = 1(t = T_B + 1)$, with critical values that depend on the break fraction $\lambda = T_B/T$. Allowing for the break, he rejected the unit root for most of the Nelson–Plosser series, reversing their conclusion.

In the original paper. Perron [143] defined three models: A, the "crash" model (a one-time change in level); B, the "changing growth" model (a change in slope); and C, both. Under the null the series is a unit root process with a one-time jump at T_B; under the alternative it is trend-stationary with a broken trend. He first showed that if the truth is trend-stationary with a break, the standard ADF estimate of ρ is biased towards one, so the test can hardly reject: a break is mistaken for persistence. Treating the 1929 crash as exogenous for the annual Nelson–Plosser series (model A) and the 1973 oil shock for post-war quarterly real GNP (model B), he rejected the unit root for 11 of the 14 series, the exceptions being consumer prices, velocity and the interest rate.

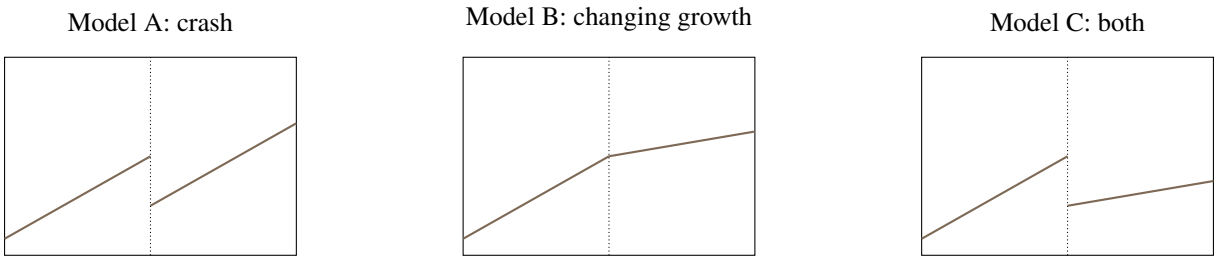


Figure 4.1: Perron's three trend-break models. The dotted line is the break date T_B. A: the level drops (the 1929 crash); B: the growth rate changes (the 1973 productivity slowdown); C: both.

4.2 ZIVOT–ANDREWS (1992)

Choosing the break after looking at the data biases Perron's test towards rejection. Zivot and Andrews [190] make the break *endogenous*: estimate the ADF regression with break dummies for every candidate date and take the most negative t-statistic, which has its own (more negative) critical values. Under the null there is a unit root *without* a break. Using this test, they found less evidence against the unit root than Perron.

R EXAMPLE: the Nile. Annual flow of the Nile at Aswan, 1871–1970, is a classic change-point series [196]. Searching over all break years for a level shift, the Zivot–Andrews statistic is minimised at -6.79 with the break after 1898 (5% critical value -4.80). Mean flow drops from 1,098 to 850 (in 10^8 m^3): the date coincides with the construction of the first Aswan dam and a change in rainfall regime. Here a plain ADF test also rejects ($t = -4.05$), but the break test locates the cause.

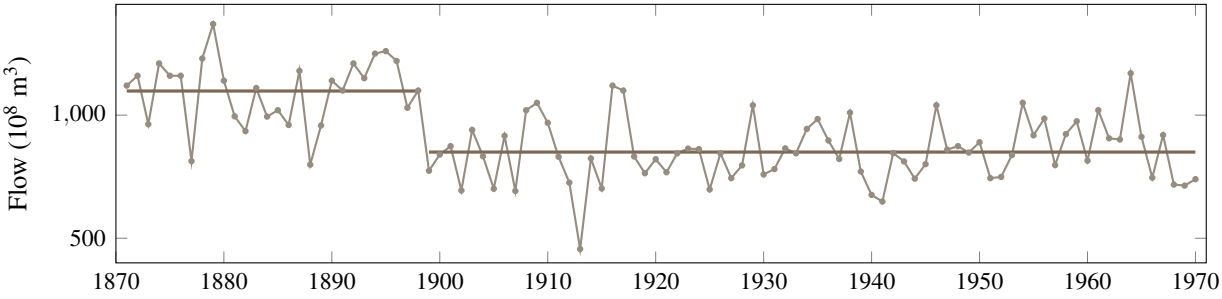


Figure 4.2: Annual Nile flow with the means before and after the break in 1898.

In the original paper. Zivot and Andrews [190] argued that Perron's break dates were chosen after looking at the data, which biases the test towards rejection (a "data-mining" or pre-test problem). They treated the break date as unknown and chose it as the date least favourable to the unit-root null, i.e. the one minimising the t-statistic, and derived the asymptotic distribution of this minimum under a null of a unit root *without* a break. With the correct, more negative critical values, they found weaker evidence against the unit root than Perron for several Nelson–

Plosser series.

4.3 LUMSDAINE–PAPELL (1997)

Extends Zivot–Andrews to *two* endogenous breaks in level and/or trend, searching over all pairs of dates (separated by a minimum distance) and taking the most negative ADF t-statistic [114]. Two breaks are natural for long annual series that span, say, a war and an oil shock. Like Zivot–Andrews, it allows breaks only under the alternative, so a unit root with breaks can be wrongly rejected; and the critical values are more negative than for one break, which lowers power. Lumsdaine and Papell found that allowing a second break strengthened the evidence against a unit root for several Nelson–Plosser series.

4.4 PERRON (1997)

One endogenous break, with two ways for the series to move to its new path. In the *additive-outlier* (AO) model the break happens instantly: the level jumps on the break date. In the *innovational-outlier* (IO) model the break works like any other shock and the series moves gradually to its new path, at the speed implied by its own dynamics. Perron offers several rules for choosing the break date: minimise the unit-root t-statistic, or maximise the significance of the break coefficients [144]. The AO model suits sudden policy changes (a currency devaluation); the IO model suits shocks whose effect spreads over time (an oil price rise feeding through to prices).

4.5 LEE–STRAZICICH (2003)

Zivot–Andrews and Lumsdaine–Papell assume no break under the null, so when the true process is a unit root *with* a break, they over-reject and suggest "trend-stationary with breaks" when the series is in fact a random walk. Lee and Strazicich's minimum LM test is built on the Schmidt–Phillips LM principle: the series is detrended under the null (in first differences), then the LM unit-root statistic is minimised over pairs of break dates. Its distribution does not depend on the break sizes, and breaks are allowed under both the null and the alternative [106]. A rejection therefore really means trend-stationarity with breaks.

In the original paper. Lee and Strazicich [106] showed by simulation that endogenous-break ADF tests that exclude breaks under the null reject far too often when the series has a unit root and a break, and tend to place the estimated break one period before the true one. Their minimum LM test, built on the Schmidt–Phillips LM principle, has a null distribution that does

not depend on the size of the breaks, which removes the problem.

4.6 LEE–STRAZICICH (2004)

The one-break version of the minimum LM test, with crash (level shift) and break (level and trend shift) models [107]. Use it when theory or history points to a single event; it has more power than the two-break test when only one break is present.

4.7 THE BAI–PERRON FRAMEWORK FOR MULTIPLE BREAKS

The foundation for everything with more than two breaks. For a regression with m breaks (stationary regressors), Bai and Perron estimate all break dates jointly by minimising the sum of squared residuals; dynamic programming reduces the search from about T^m to $O(T^2)$ operations. The number of breaks is chosen by sequential sup-$F(\ell+1 \mid \ell)$ tests (is there one more break?), by double-maximum tests, or by an information criterion [206, 207]. Most of the tests below use this machinery.

4.8 OHARA (1999)

An early generalisation of Zivot–Andrews to m trend breaks at unknown dates, applied to US and Japanese macroeconomic series [208].

4.9 KAPETANIOS (2005)

Unit-root test against the alternative of up to m breaks. Breaks are found one at a time: fix the first break where it reduces the sum of squared residuals most, then search for the next, and so on, which is far cheaper than a joint search. The test statistic is the smallest ADF t over the sequence [209]. As with Zivot–Andrews, breaks are allowed only under the alternative.

4.10 CARRION-I-SILVESTRE, KIM AND PERRON (2009)

The reference test for multiple breaks. It allows an arbitrary number of breaks in level and slope under *both* the null and the alternative, so a unit root with breaks is not mistaken for stationarity. The series is GLS-detrended (as in DF-GLS) and tested with the modified M-statistics of Ng and Perron, giving power close to the Gaussian power envelope [210]. Up to five breaks are

tabulated in practice.

4.11 COUNTING THE BREAKS FIRST: KEJRIWAL AND PERRON (2010)

Which break dates and how many? Testing for breaks in a trend depends on whether the noise is $I(0)$ or $I(1)$, which is the very question being asked. Kejriwal and Perron give a sequential procedure whose size is robust to either case [211]. The recommended workflow is: count and date the breaks with this procedure, then feed them to the Carrion-i-Silvestre–Kim–Perron test.

4.12 HARVEY, LEYBOURNE AND TAYLOR (2013)

Takes the infimum of local GLS-detrended DF statistics over possible break points, for multiple trend breaks. Unlike Zivot–Andrews it does not spuriously reject when large breaks are present under the null, and its power holds up for breaks of plausible size [212].

4.13 UNKNOWN NUMBER OF SMOOTH BREAKS: FOURIER TESTS

Instead of dating sharp breaks, approximate any slowly shifting mean by a few low-frequency sine and cosine terms:

$$y_t = \alpha + \gamma_1 \sin\!\left(\frac{2\pi k t}{T}\right) + \gamma_2 \cos\!\left(\frac{2\pi k t}{T}\right) + \text{error}, \tag{4.1}$$

with a small frequency k (often 1 or 2). The approach needs no break dates and no prior number of breaks, and suits gradual change (deregulation phased in, slow technological shifts). Becker, Enders and Lee give a KPSS-type stationarity test [213]; Enders and Lee give LM and Dickey–Fuller-type unit-root versions [214, 215]. Sharp, one-off events are still better handled with dated dummies.

Test	Breaks	Breaks under the null?
Zivot–Andrews; Lumsdaine–Papell	1; 2	No
Lee–Strazicich	1–2	Yes
Kapetanios (2005)	up to m	No
Carrion-i-Silvestre–Kim–Perron (2009)	any m (usually ≤ 5)	Yes
Harvey–Leybourne–Taylor (2013)	multiple	Robust to breaks under the null
Fourier (Enders–Lee; Becker–Enders–Lee)	unknown number, smooth	Yes

Table 4.1: Unit-root tests by the number of breaks they allow.

A SIMPLE EXAMPLE. A series is $10, 11, 9, 10, 20, 21, 19, 20$: noise around 10, then noise around 20 after a policy change. Shocks inside each half die out at once, so the series is stationary around a shifted mean. An ADF test without a break sees the level move from 10 to 20 and never come back, and reads it as a permanent shock (a unit root). Adding a level-shift dummy $DU_t = 1(t > 4)$ absorbs the jump; the remaining residuals $0, 1, -1, 0, 0, 1, -1, 0$ are plain noise, and the unit root is rejected.

> IN PLAIN WORDS. Before declaring that shocks are permanent, ask whether a few special events (a war, a crash, a policy change) explain the apparent drift. Break tests let the data choose those events: count the breaks first, then test for a unit root allowing for them.

* * *

CHAPTER 5

COINTEGRATION TESTS WITHOUT BREAKS

Variables that are $I(1)$ individually are *cointegrated* if a linear combination $z_t = y_t - \beta' \mathbf{x}_t$ is $I(0)$: they wander, but together. They then share a long-run equilibrium, and by the Granger representation theorem they have an error-correction representation [44]:

$$\Delta y_t = \alpha\,(y_{t-1} - \beta' \mathbf{x}_{t-1}) + \text{lags of } \Delta y, \Delta \mathbf{x} + \varepsilon_t, \qquad \alpha < 0. \tag{5.1}$$

Analogy. A drunk walking a dog: each wanders randomly, but the lead keeps them close; α is how hard the lead pulls.

Real-world pairs. Spot and futures prices of the same commodity; consumption and income; stock prices and dividends; short and long interest rates; prices in two countries and their exchange rate (PPP).

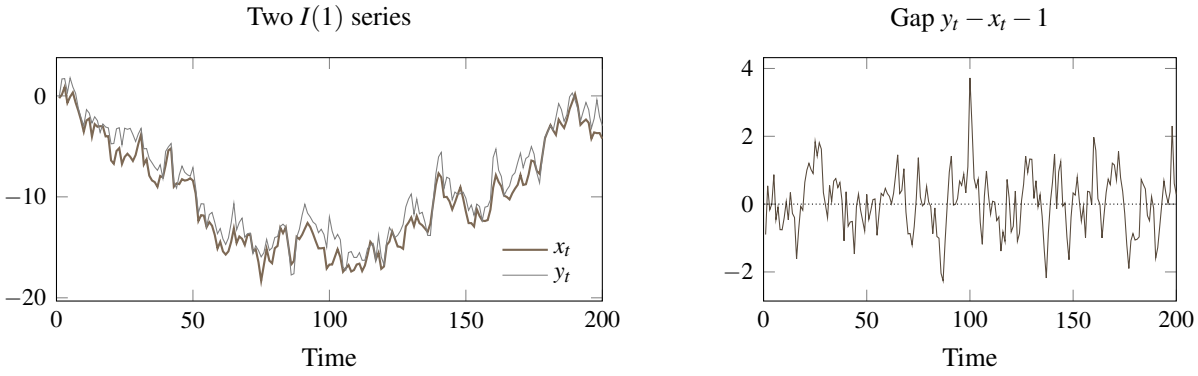


Figure 5.1: Cointegration in a picture. Left: both series wander without a fixed mean. Right: their gap keeps returning to zero, the long-run equilibrium. Simulated data.

5.1 COINTEGRATING REGRESSION DURBIN–WATSON (CRDW)

The Durbin–Watson statistic of the levels regression $y_t = \mu + \beta x_t + z_t$. If the residual z_t is a random walk, consecutive residuals are almost equal and $DW = \sum(\hat{z}_t - \hat{z}_{t-1})^2 / \sum \hat{z}_t^2 \to 0$; if it is stationary, DW is bounded away from zero. So H_0: DW $= 0$ (no cointegration), rejected for large DW [158]. A quick screen, also useful as a warning sign: a regression of two trending series with R^2 above DW is a classic symptom of spurious regression. Its critical values depend on the dynamics of the data, so it is rarely used alone.

5.2 ENGLE–GRANGER (1987)

Two steps [44]: (1) estimate $y_t = \mu + \beta' \mathbf{x}_t + z_t$ by OLS (super-consistent: $\hat{\beta}$ converges at rate T); (2) run an ADF test on $\hat{z}_t$ with Engle–Granger/MacKinnon critical values (about -3.34 at 5% for two variables with a constant), which are more negative than DF values because $\hat{\beta}$ was chosen to make $\hat{z}_t$ look stationary.

R EXAMPLE: DAX and CAC. Regressing log DAX on log CAC (daily, 1991–98) gives a slope of 1.55, but the ADF t on the residuals is only -2.03: no cointegration. Two stock markets can be highly correlated in returns and still drift apart in levels.

In the original paper. Engle and Granger [44] defined variables as cointegrated of order (d,b) when each is $I(d)$ but a combination is $I(d-b)$, and proved the Granger representation theorem: a cointegrated system has equivalent VAR, moving-average and error-correction representations. Two implications followed: a VAR in first differences is misspecified for cointegrated data (it omits the error-correction term), and a levels regression estimates β consistently at rate T. They compared seven candidate test statistics by Monte Carlo, including the CRDW, Dickey–Fuller and augmented Dickey–Fuller tests on the residuals, and recommended the ADF version. They applied the methods to several US examples such as consumption and income and short- and long-term interest rates.

5.3 JOHANSEN (1991)

The system approach. Write the VAR in levels as a VECM: $\Delta \mathbf{y}_t = \Pi \mathbf{y}_{t-1} + \sum_i \Gamma_i \Delta \mathbf{y}_{t-i} + \varepsilon_t$. If $\mathbf{y}_t$ is $I(1)$, the rank r of Π is the number of cointegrating vectors, and $\Pi = \alpha \beta'$ (β: long-run relations, α: adjustment speeds). The rank is found by reduced-rank regression with eigenvalues

$\hat\lambda_1 > \cdots > \hat\lambda_n$ [91, 92]:

$$\text{trace: } -T\sum_{i=r+1}^{n}\ln(1-\hat\lambda_i), \qquad \text{max-eigenvalue: } -T\ln(1-\hat\lambda_{r+1}), \tag{5.2}$$

tested sequentially for $r = 0, 1, \ldots$. Advantages over Engle–Granger: more than one cointegrating vector, no need to choose which variable to normalise on, and tests of restrictions on α and β.

Application. Johansen and Juselius [197] found one cointegrating vector among money, income, prices and interest rates for Denmark and Finland: a stable long-run money demand function.

In the original paper. Johansen [91, 92] derived maximum-likelihood estimation of the cointegrating space in a Gaussian VAR. After concentrating out the short-run dynamics, the problem reduces to a *reduced-rank regression*: the eigenvalues $\hat\lambda_i$ are the squared canonical correlations between $\Delta\mathbf{y}_t$ and $\mathbf{y}_{t-1}$, and the corresponding eigenvectors estimate β. The likelihood-ratio statistic for the rank has a limiting distribution given by functionals of Brownian motion (tabulated by simulation), which depends on how constants and trends enter; linear restrictions on β and α can then be tested with ordinary χ^2 tests.

5.4 PESARAN–SHIN–SMITH BOUNDS TESTING

Estimate an unrestricted error-correction (ARDL) model $\Delta y_t = c + \pi_y y_{t-1} + \pi_x' \mathbf{x}_{t-1} + \text{lags} + \varepsilon_t$ and F-test $\pi_y = \pi_x = \mathbf{0}$ [146, 148]. Two sets of critical values bound the cases where all regressors are $I(0)$ (lower bound) and all $I(1)$ (upper bound): above the upper bound, a level relation exists whatever the integration order; below the lower, it does not; in between, the test is inconclusive. No prior unit-root classification is needed, but $I(2)$ series are ruled out. The original paper applied it to a UK earnings equation.

In the original paper. Pesaran, Shin and Smith [148] asked whether a long-run level relationship can be tested without first knowing whether the regressors are $I(0)$ or $I(1)$, since unit-root pre-tests have low power. In a conditional error-correction model they derived the asymptotic distributions of the Wald (F) and t statistics on the lagged levels under both extreme assumptions, giving two sets of critical values that bound all intermediate cases, for five specifications of the deterministic terms. Their application was a UK earnings equation relating real wages to productivity, unemployment, union density and the tax wedge.

5.5 NARAYAN (2005)

Small-sample critical values for the bounds test for roughly 30–80 observations, with an application to saving and investment in China [128]. Annual macro data for a single emerging economy are typically this short, which is why the paper is widely cited.

5.6 NULL OF COINTEGRATION: SHIN (1994)

A KPSS-type LM test on the residuals of an efficiently estimated cointegrating regression, with cointegration as the *null* [225]. Used alongside Engle–Granger for confirmatory analysis, as KPSS is used alongside ADF.

5.7 NONLINEAR ARDL (NARDL)

Shin, Yu and Greenwood-Nimmo split a regressor into its cumulative increases and decreases, $x_t^+ = \sum_{j\le t} \max(\Delta x_j, 0)$ and $x_t^- = \sum_{j\le t} \min(\Delta x_j, 0)$, and put both into the ARDL bounds framework [226]. Long- and short-run asymmetry can then be tested directly. The classic application is "rockets and feathers": retail fuel prices rise quickly when crude oil prices rise but fall slowly when they fall.

5.8 EXACT CRITICAL VALUES FOR THE BOUNDS TEST

Kripfganz and Schneider estimated response-surface regressions that give bounds critical values and approximate p-values for any sample size and number of regressors [227], replacing interpolation in the Pesaran–Shin–Smith and Narayan tables.

A SIMPLE EXAMPLE. Futures and spot prices of gold satisfy, in equilibrium, $F_t = S_t + 2$ (the cost of carry). Today $S = 100$, $F = 103$: the gap from equilibrium is $z = 103 - 100 - 2 = 1$. In the error-correction model $\Delta F_{t+1} = -0.5\, z_t + \cdots$, the futures price is expected to fall by 0.5 tomorrow, closing half the gap. Each price on its own wanders like a random walk, but the gap z_t keeps returning to zero: they are cointegrated with $\beta = (1, -1)$.

> IN PLAIN WORDS. Two wandering series are cointegrated if they never wander too far apart. Finding that lead (the long-run relation) and how fast it pulls (the adjustment speed) is the purpose of these tests.

* * *

Chapter 6

Cointegration Tests with Breaks

A shift in the long-run relation (a new monetary regime, a currency peg, a change in taxation) makes the residual of a fixed relation look non-stationary, so residual-based and rank tests wrongly fail to find cointegration. These tests allow for shifts.

6.1 Gregory–Hansen (1996)

Residual-based test with one unknown break in the intercept (level shift), intercept and trend, or the full coefficient vector (regime shift): $y_t = \mu_1 + \mu_2 D_{t\tau} + \beta_1' \mathbf{x}_t + \beta_2' \mathbf{x}_t D_{t\tau} + z_t$, $D_{t\tau} = 1(t > \tau T)$. The statistics ADF^*, Z_t^*, Z_α^* are the infimum over break dates [66]. The null is no cointegration; a rejection means cointegration *possibly* with a break, not proof of a break.

In the original paper. Gregory and Hansen [66] showed that conventional residual-based tests lose power when the cointegrating vector shifts, because the residual then contains a step. They defined three models (level shift; level shift with trend; regime shift in both intercept and slopes), computed the ADF, Z_α and Z_t statistics for each break date in the middle 70% of the sample, took the smallest, and derived its limiting distribution. They stress that a rejection supports cointegration; it does not by itself establish that a break occurred, which should be tested separately.

6.2 Johansen–Mosconi–Nielsen

Johansen's likelihood-ratio rank test with breaks in the deterministic terms (level and trend) at *known* dates. The sample is divided into q sub-periods; each has its own intercept and trend, the trend coefficients are restricted to lie in the cointegrating space, and dummies for the first observations after each break remove their influence [94]. The asymptotic distribution depends on the relative positions of the breaks, and the authors provide response-surface critical values.

Typical uses: German data around reunification, euro-area data around 1999, and any system with a known regime change.

6.3 CARRION-I-SILVESTRE AND SANSÓ (2006)

A KPSS-type LM test with the null hypothesis of *cointegration*, allowing a structural break at an unknown date in the deterministic terms and/or the cointegrating vector [27]. The regression is estimated by dynamic OLS to handle endogeneity, and the break date is chosen by minimising the sum of squared residuals. It is the confirmatory counterpart of Gregory–Hansen: Gregory–Hansen rejecting and Carrion-i-Silvestre–Sansó not rejecting is strong evidence for cointegration with a break.

6.4 LÜTKEPOHL, SAIKKONEN AND TRENKLER (2004)

A rank test for a VAR with a level shift at an *unknown* date. Step one estimates the shift date from a VAR in first differences (where a level shift becomes a single impulse); step two estimates and removes the deterministic part, including the shift, by GLS; step three applies a Johansen-type test to the adjusted series. The limiting distribution does not depend on the break date [118], so standard tables apply.

6.5 SAIKKONEN AND LÜTKEPOHL (2000)

Rank test for a VAR with shifts in the deterministic term at *known* dates. The deterministic part is estimated by GLS and subtracted, and a likelihood-ratio-type test is applied to the adjusted series; the critical values do not depend on the break dates [157]. German reunification in 1990 is the standard application: a level shift in German macro series at a known date.

6.6 INOUE (1999)

Rank tests when the deterministic trend has a break at an unknown date. Inoue generalises the Johansen trace and maximum-eigenvalue statistics by taking the infimum over candidate break dates, with new critical values [89]. Useful for long macro systems where productivity growth slowed at an uncertain date (such as the early 1970s).

6.7 HATEMI-J (2008)

Extends Gregory–Hansen to two unknown regime shifts: both the intercept and the slope coefficients can change twice. The break dates are found by a two-dimensional grid search that minimises the ADF (or Z_t, Z_α) statistic of the residuals, and new critical values are tabulated for up to four regressors [80]. The paper applies it to financial market integration, where both liberalisation and crises can shift the relation.

6.8 NARAYAN AND POPP

Listed in the outline as "Narayan and Pope (2016)". The exact paper could not be matched; the closely related Narayan–Popp test is a *unit-root* test with two breaks in level and slope [129]. Please confirm the intended reference.

6.9 SEVERAL KNOWN BREAKS: JOHANSEN–MOSCONI–NIELSEN

Item 6.2 already allows more than two breaks: the trend may shift at several known dates, splitting the sample into q sub-periods, and the critical values depend on where the breaks fall [94].

6.10 MULTIPLE BREAKS IN COINTEGRATED REGRESSIONS: KEJRIWAL AND PERRON

Bai–Perron for cointegrated regressions. Kejriwal and Perron derive the limiting distribution of the break-date and coefficient estimates when an $I(1)$ cointegrating relation has several breaks [216], and give sup-Wald and sequential tests for the number of breaks in the long-run coefficients, allowing some coefficients to stay fixed [217]. They answer "has the long-run relation changed, how many times, and when?", for example whether the loss-rate–unemployment relation in a credit model held across 2008 and 2020.

6.11 MAKI (2012)

Extends Gregory–Hansen (one break) and Hatemi-J (two breaks) to an unknown number of breaks, up to a user-chosen maximum (often five). The null is no cointegration; breaks are found sequentially, and four specifications allow level shifts or regime shifts, with or without

trend [218]. Implemented in the R package `makicoint`.

6.12 FOURIER COINTEGRATION TESTS

Smooth breaks of unknown number in the cointegrating relation, approximated by Fourier terms: Tsong, Lee, Tsai and Hu test the null of cointegration [219]; Banerjee, Arčabić and Lee give an ARDL-based test with the null of no cointegration and apply it to crude oil prices [220].

A SIMPLE EXAMPLE. Suppose the cost of carry rises from 2 to 5 after an interest-rate increase halfway through the sample, so $F_t = S_t + 2$ before and $F_t = S_t + 5$ after. A regression with one fixed intercept estimates about 3.5 and leaves residuals near -1.5 in the first half and $+1.5$ in the second: a step that never reverses, which an Engle–Granger test reads as a non-stationary residual, wrongly concluding "no cointegration". A Gregory–Hansen level-shift dummy lets the intercept change and recovers the relation.

> IN PLAIN WORDS. If the lead between the drunk and the dog was lengthened once, a test that assumes a fixed lead will say there is no lead at all. These tests allow the lead to change.

* * *

Chapter 7

Causality Tests

7.1 Granger causality

Idea. x "Granger-causes" y if the past of x helps forecast y beyond what the past of y already tells us. In $y_t = a + \sum_{j=1}^{p} a_j y_{t-j} + \sum_{j=1}^{q} b_j x_{t-j} + u_t$ test H_0: $b_1 = \cdots = b_q = 0$ with an F or Wald test [61]. It measures *predictive precedence*, not structural causation: the weather forecast Granger-causes rain. The variables must be stationary (or the test adapted, Section 7.4), and omitted third variables can create spurious causality.

Application. Sims [162] found that money Granger-caused nominal income in post-war US data but not the reverse, a key piece of evidence in the monetarist debate.

R example. For daily DAX and FTSE returns with two lags, the test of "FTSE does not Granger-cause DAX" has $F = 2.09$ ($p = 0.12$), and "DAX does not Granger-cause FTSE" has $F = 3.30$ ($p = 0.037$): weak evidence that German news leads London by a day.

In the original paper. Granger [61] defined causality through predictability: X causes Y if Y can be predicted better using all available information than using all information except the past of X. He also defined *feedback* (causality both ways), *instantaneous causality* (current X improves the prediction of current Y) and the causality lag, and showed how cross-spectral methods measure the strength of causality frequency by frequency. He warned that omitted variables and temporal aggregation can create apparent causality, and that the definition only makes sense for stochastic, stationary series.

7.2 Sims causality

Regress y_t on past, current *and future* values of x: $x_t = \sum_{j=-q}^{p} c_j y_{t-j} + v_t$. If x does not respond to y (so that y does not Granger-cause x), the coefficients on future y are zero [162]. The test is

the mirror image of Granger's and gives the same answer in population; Sims used it to show that money was not caused by income in US data, supporting money's role as exogenous.

In the original paper. Sims [162] proved that Y fails to Granger-cause X if and only if, in a regression of Y on past, current and future X, all coefficients on future X are zero. He pre-filtered the data to reduce serial correlation and applied the test to quarterly US money (M1 and the monetary base) and nominal GNP. In regressions of GNP on past, current and future money, the future money terms were insignificant, so GNP did not cause money; in regressions of money on past, current and future GNP, future GNP mattered. He read this as one-way causality from money to income.

7.3 CAUSALITY THROUGH WEAK AND STRONG EXOGENEITY

Engle, Hendry and Richard [45] distinguish three levels: *weak exogeneity* of x for the parameters of interest (we can model y given x without losing information); *strong exogeneity* (weak exogeneity plus y does not Granger-cause x; needed for forecasting y with forecasts of x); and *super exogeneity* (weak exogeneity plus invariance of the conditional model to changes in the process of x, e.g. policy changes; needed for policy analysis and the answer to the Lucas critique). In a VECM, x is weakly exogenous for β if its adjustment coefficient in α is zero.

7.4 TODA–YAMAMOTO (1995)

When series may be $I(1)$ or cointegrated, Wald tests in a levels VAR have non-standard distributions. Toda and Yamamoto fit a VAR($k+d_{\max}$) in levels, where $d_{\max}$ is the highest suspected order of integration, and Wald-test only the first k lags; the extra lags absorb the non-standard part, so the statistic is asymptotically χ^2_k whether the series are $I(0)$, $I(1)$ or cointegrated [172]. No pre-testing is needed, at some cost in power. It is the standard tool in the energy-consumption and economic-growth literature, where the integration properties of short annual series are uncertain.

In the original paper. Toda and Yamamoto [172] built on the result that Wald tests of restrictions in a levels VAR with integrated variables have non-standard limiting distributions that depend on nuisance parameters. Adding $d_{\max}$ extra lags that are not tested turns the restricted coefficients into parameters whose estimators have standard asymptotics. They noted the cost: the extra lags waste degrees of freedom, so the test is less efficient than one that exploits known cointegration, and the lag length k must be chosen correctly first.

7.5 CAUSALITY IN THE FREQUENCY DOMAIN

Granger causality may hold at some cycle lengths and not others: money may help forecast prices in the long run but not month to month. Breitung and Candelon decompose the test by frequency, testing linear restrictions on the lag coefficients at each frequency [228].

7.6 PANEL GRANGER CAUSALITY

Dumitrescu and Hurlin run Granger tests unit by unit and average the Wald statistics, allowing the coefficients to differ across units; the null is no causality in any unit [229]. Widely used for energy–growth and finance–growth panels.

A SIMPLE EXAMPLE. Does advertising (x) help forecast sales (y)? With 104 quarterly observations, regress sales on two of its own lags: $\mathrm{SSR}_R = 120$. Add two lags of advertising: $\mathrm{SSR}_U = 100$, with 5 estimated coefficients. Then

$$F = \frac{(120 - 100)/2}{100/(104 - 5)} = \frac{10}{1.01} = 9.9,$$

far above the 5% critical value of $F(2, 99) \approx 3.09$: advertising Granger-causes sales. It does not prove that advertising causes sales; firms may advertise ahead of seasons they already expect to be strong.

> IN PLAIN WORDS. Granger causality asks a modest question: does knowing x's past improve my forecast of y? It cannot tell you what would happen if you changed x; that needs a structural model.

* * *

Chapter 8

Vector Autoregressive Model (VAR)

A VAR(p) is $\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \varepsilon_t$ with $\mathrm{E}[\varepsilon_t\varepsilon_t'] = \Sigma$ [117, 163]. Every variable is explained by the past of every variable, so no variable has to be declared exogenous in advance; each equation can be estimated by OLS. The VAR is stable if all eigenvalues of the companion matrix lie inside the unit circle. With K variables and p lags there are $K(Kp+1)$ coefficients, which is why large VARs need shrinkage (Chapter 33).

Sims' 1980 critique of large structural macro models ("incredible" identifying restrictions) introduced the VAR as the workhorse of empirical macroeconomics; a three-variable VAR in inflation, unemployment and the interest rate is the textbook example [199].

8.1 Impulse response function (IRF)

The moving-average form $\mathbf{y}_t = \mu + \sum_{s\ge 0}\Psi_s\varepsilon_{t-s}$ (with $\Psi_0 = \mathbf{I}$, $\Psi_s = \sum_{j=1}^{s}\Psi_{s-j}\mathbf{A}_j$) traces the response of $\mathbf{y}_{t+s}$ to a shock today. Because reduced-form errors are correlated, shocks are orthogonalised with the Cholesky factor $\Sigma = \mathbf{P}\mathbf{P}'$: responses $\Psi_s\mathbf{P}$. This imposes a recursive ordering: the first variable is not affected contemporaneously by the others, so results can depend on the ordering. Generalised IRFs integrate out the other shocks using the historical correlation and are invariant to ordering [100, 145]. Confidence bands come from bootstrap or asymptotic formulas.

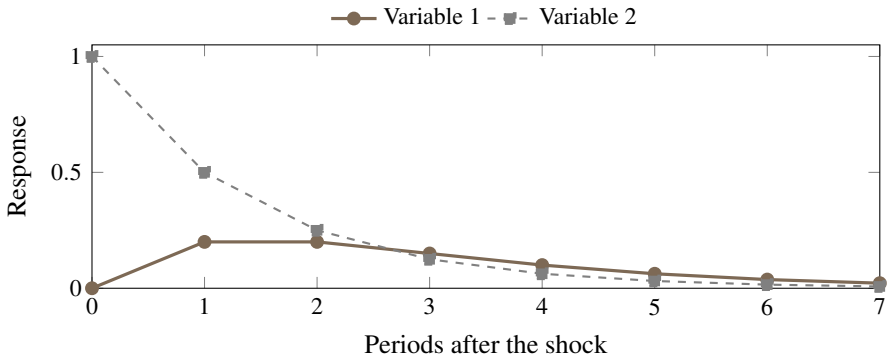


Figure 8.1: Impulse responses to a unit shock in variable 2 in the VAR(1) of the simple example. The shock decays in its own variable and spills over, with a delay and a hump, into variable 1.

8.2 VARIANCE DECOMPOSITION

The forecast-error variance decomposition (FEVD) answers "what share of the h-step forecast error variance of variable i is due to shock j?": $\omega_{ij}(h) = \sum_{s=0}^{h-1}(\mathbf{e}_i'\Psi_s\mathbf{p}_j)^2 / \sum_{s=0}^{h-1}\mathbf{e}_i'\Psi_s\Sigma\Psi_s'\mathbf{e}_i$ [117]. Shares sum to one across shocks for each variable and horizon.

Application. Monetary-policy VARs typically find that policy shocks explain only a small share of output fluctuations at business-cycle horizons: most of the variation in the policy rate is a systematic response to the economy, not a surprise [200].

8.3 LOCAL PROJECTIONS

Instead of iterating a VAR forward, Jordà estimates the impulse response at each horizon h by a separate regression, $y_{t+h} = \alpha_h + \beta_h\,\text{shock}_t + \text{controls}_t + u_{t+h}$ [230]. Local projections are robust to misspecification of the VAR dynamics and easy to extend (state dependence, panels), at the cost of more variable estimates. Plagborg-Møller and Wolf show that, with unrestricted lags, local projections and VARs estimate the same population impulse responses; they differ in the bias–variance trade-off in finite samples [231].

A SIMPLE EXAMPLE. A two-variable VAR(1), $\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t$, with $\mathbf{A} = \left(\begin{smallmatrix} 0.5 & 0.2 \\ 0 & 0.5 \end{smallmatrix}\right)$: variable 2 affects variable 1, not the reverse. A shock of 1 to variable 2 gives the responses

Period	0	1	2	3
Variable 1	0	0.20	0.20	0.15
Variable 2	1	0.50	0.25	0.125

computed as $\mathbf{A}(0,1)' = (0.2, 0.5)'$, then $\mathbf{A}(0.2, 0.5)' = (0.2, 0.25)'$, and so on: the shock spills

over into variable 1 with a delay, peaks, and fades.

> IN PLAIN WORDS. A VAR lets every variable talk to every other variable through their pasts. The impulse response is the echo of one shock travelling through the system over time.

* * *

CHAPTER 9

VECTOR ERROR CORRECTION MODEL (VECM)

For cointegrated $I(1)$ systems, $\Delta \mathbf{y}_t = \alpha \beta' \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \varepsilon_t$, where β holds the r cointegrating vectors (the long run) and α the speeds of adjustment back to equilibrium [92]. A VAR in differences would throw away the long-run information; a VAR in levels keeps it but ignores the rank restriction.

Reading α. In a VECM for spot (s) and futures (f) prices with $\beta = (1, -1)'$, an α_s significantly negative and an α_f close to zero mean the spot price does the adjusting: the futures market leads in price discovery.

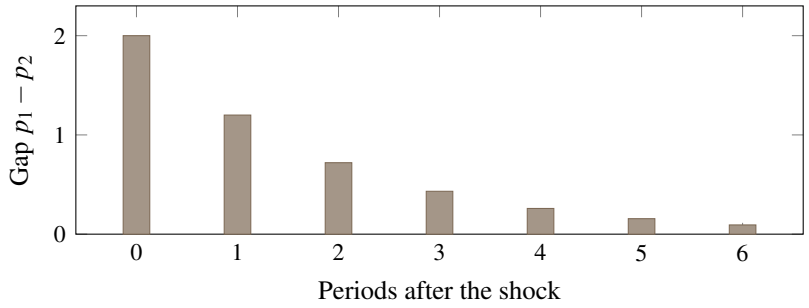


Figure 9.1: Error correction at work: with $\alpha = (-0.3, 0.1)$, a gap of 2 between two cointegrated prices shrinks by 40% each period $(2, 1.2, 0.72, \ldots)$, as in the simple example of this chapter.

9.1 FORECAST-ERROR IMPULSE RESPONSE (FEIRF) FOR THE VECM

In a cointegrated system, shocks to the common stochastic trends have permanent effects, so impulse responses converge to non-zero constants rather than dying out [119, 145]. The long-run

response is **CP** (Chapter 11), which has rank $K - r$. Responses of the cointegrating combinations $\beta' \mathbf{y}_t$ do die out; their profile, the *persistence profile*, shows how fast equilibrium is restored after a system-wide shock and does not depend on how shocks are identified [145]. Confidence intervals are best obtained by bootstrap, since the asymptotic distributions are non-standard when responses do not decay.

A SIMPLE EXAMPLE. Two prices should move together, $\beta = (1, -1)$, with adjustment speeds $\alpha = (-0.3, 0.1)$. Today the gap $p_1 - p_2 = 2$. Next period $\Delta p_1 = -0.3 \times 2 = -0.6$ and $\Delta p_2 = 0.1 \times 2 = +0.2$; the gap becomes $2 - 0.6 - 0.2 = 1.2$. Price 1 does three-quarters of the adjusting; price 2 does the rest.

> IN PLAIN WORDS. A VECM is a VAR with a memory of where the system should be. When the variables stray from their long-run relation, the error-correction term pulls them back.

* * *

CHAPTER 10

STRUCTURAL VAR (SVAR)

Reduced-form errors ε_t are mixtures of economically meaningful structural shocks $\mathbf{u}_t$ (monetary policy, demand, supply, oil), with $\mathbf{A}\varepsilon_t = \mathbf{B}\mathbf{u}_t$ and $\mathrm{E}[\mathbf{u}_t\mathbf{u}_t'] = \mathbf{I}$ [117]. The data give only Σ, which has $K(K+1)/2$ distinct elements; if $\mathbf{A} = \mathbf{I}$ the K^2 elements of $\mathbf{B}$ need $K(K-1)/2$ further restrictions. Different identification schemes supply them.

10.1 RECURSIVE IDENTIFICATION (SIMS, 1980)

$\mathbf{B}$ lower triangular (the Cholesky factor): variable 1 responds to no other shock within the period, variable 2 only to shock 1, and so on [163]. In monetary VARs, output and prices are ordered before the policy rate: the central bank sees current activity, but activity reacts to policy only with a lag.

10.2 WOLD RESTRICTION (CAUSAL CHAIN)

The same recursive structure justified by timing within the period (a Wold causal chain): e.g. with monthly data, production plans are set before prices and interest rates are observed [117].

10.3 IDENTIFICATION FROM ECONOMIC THEORY

Non-recursive contemporaneous restrictions from theory, such as a money-demand and a money-supply equation [12, 164]; and long-run restrictions, the best known being Blanchard and Quah's: demand shocks have no permanent effect on output, supply shocks may [16]. Applied to US output and unemployment, demand shocks explained a large share of short-run fluctuations and supply shocks the long-run level of output.

Application. Kilian identified oil supply shocks, global demand shocks and oil-specific demand shocks and showed that their macroeconomic effects differ sharply: an oil price rise driven by booming world demand is not the same event as one driven by a supply disruption [201].

10.4 IDENTIFICATION THROUGH HETEROSKEDASTICITY

If the variances of the structural shocks change across regimes (calm and crisis periods) while **B** does not, the extra moments identify **B** without zero restrictions [155]. Rigobon applied it to contagion among emerging-market sovereign bond markets.

10.5 SIGN RESTRICTIONS

Instead of zeros, impose the *signs* of some responses for a few periods (a contractionary monetary shock raises the interest rate and lowers prices), draw many rotations **PQ** with **Q** orthogonal, and keep those that satisfy the signs [49, 178]. The result is a *set* of admissible models. Uhlig found that, leaving the output response unrestricted, a contractionary monetary shock has no clear effect on real output.

10.6 RESTRICTIONS FROM A DSGE MODEL

Use the restrictions implied by a dynamic stochastic general equilibrium model, or check whether the DSGE model has a VAR representation at all (the "ABCs and Ds" condition of [50]); estimated DSGE models such as Smets–Wouters are used by central banks for forecasting and policy analysis [166].

10.7 EXTERNAL INSTRUMENTS (PROXY SVAR)

Use an outside measure correlated with the shock of interest and uncorrelated with the others, such as a narrative series of tax changes or high-frequency surprises in interest-rate futures around central-bank announcements, as an instrument for the structural shock [232, 233]. Mertens and Ravn used narrative tax changes to estimate the effects of US personal and corporate income tax cuts. This is now the most common identification strategy for monetary policy shocks.

10.8 NARRATIVE SIGN RESTRICTIONS

Sign restrictions often leave the admissible set too wide. Antolín-Díaz and Rubio-Ramírez add restrictions on the shocks at specific historical dates, e.g. that the monetary shock in October 1979 (the Volcker tightening) was contractionary and the largest contributor to the interest-rate change that month [234]. A handful of such restrictions sharpens inference considerably.

A SIMPLE EXAMPLE. Reduced-form errors in output (e_1) and the interest rate (e_2) have covariance matrix $\Sigma = \left(\begin{smallmatrix} 1 & 0.5 \\ 0.5 & 1.25 \end{smallmatrix}\right)$. Its Cholesky factor is $\mathbf{P} = \left(\begin{smallmatrix} 1 & 0 \\ 0.5 & 1 \end{smallmatrix}\right)$ (check: $\mathbf{P}\mathbf{P}' = \Sigma$). Ordering output first means: a demand shock of 1 raises output by 1 and the interest rate by 0.5 in the same quarter (the central bank reacts), while a monetary shock of 1 raises the rate by 1 and leaves output unchanged on impact. Reverse the ordering and the story, and the numbers, change: that is why the identifying assumption must be defended.

> IN PLAIN WORDS. A VAR sees only the mixed-up surprises in the data. An SVAR unmixes them into named shocks, but only by adding assumptions, and every conclusion is as credible as those assumptions.

* * *

Chapter 11

Structural Vector Error Correction Model (SVECM)

Idea. In a cointegrated system some shocks leave a permanent mark (a productivity improvement raises output for ever) and some fade away (a burst of demand raises output for a while). A structural VECM (SVECM) uses the cointegration structure to tell these two kinds of shock apart and gives them economic names [93, 99].

11.1 The common-trends representation

For a K-variable system with r cointegrating vectors, the Granger–Johansen representation writes the levels as

$$\mathbf{y}_t = \mathbf{C}\sum_{s\le t}\boldsymbol{\varepsilon}_s + \mathbf{C}^*(L)\boldsymbol{\varepsilon}_t + \mathbf{y}_0^*, \qquad \mathbf{C} = \boldsymbol{\beta}_\perp\Big(\boldsymbol{\alpha}_\perp'\Big(\mathbf{I}-\sum_i \boldsymbol{\Gamma}_i\Big)\boldsymbol{\beta}_\perp\Big)^{-1}\boldsymbol{\alpha}_\perp'.$$

The first term is a sum of all past shocks, a random walk: it holds the $K-r$ *common stochastic trends*. The second term is stationary. The matrix $\mathbf{C}$ has rank $K-r$, so only $K-r$ combinations of shocks can have permanent effects; the other r shocks are *transitory*.

11.2 Identification

With structural shocks $\mathbf{u}_t = \mathbf{B}^{-1}\boldsymbol{\varepsilon}_t$, the long-run impact matrix is $\Xi\mathbf{B}$ (with $\Xi = \mathbf{C}$) and the short-run impact matrix is $\mathbf{B}$. Because $\Xi\mathbf{B}$ has only $K-r$ non-zero columns, the r transitory shocks are identified as the ones with zero long-run effect; this supplies $r(K-r)$ restrictions for free. A further $(K-r)(K-r-1)/2$ restrictions separate the permanent shocks from each other and $r(r-1)/2$ the transitory ones, typically as zeros in $\Xi\mathbf{B}$ or $\mathbf{B}$ [117]. Estimation is by

maximum likelihood on the VECM, then solving for $\mathbf{B}$.

11.3 APPLICATIONS

King, Plosser, Stock and Watson found that US output, consumption and investment share a single common stochastic trend, interpreted as productivity, and that permanent shocks explain a large share, but not all, of post-war fluctuations [99]. Johansen and Juselius identified long-run and short-run structure in an IS–LM system for the UK [93]. In finance, a system of bond yields typically has one common trend (the level of rates) and transitory spread shocks.

A SIMPLE EXAMPLE. Output and consumption share one productivity trend ($K = 2$, $r = 1$, so one permanent and one transitory shock). A productivity shock raises both by 1% for ever. A demand shock raises output by 0.5% on impact, 0.25% a year later, and nothing in the long run. The long-run restriction "the demand shock has zero permanent effect on output" is exactly what separates the two shocks.

* * *

CHAPTER 12

STRUCTURAL COINTEGRATING VAR (S-CI-VAR)

Idea. Let economic theory decide where the economy must end up in the long run, and let the data decide how it gets there. A structural cointegrating VAR imposes theory-based restrictions on the cointegrating vectors β, tests them, and then models the short run freely [136, 147].

12.1 LONG-RUN STRUCTURAL MODELLING

Pesaran and Shin's approach for a small open economy lists the long-run relations that theory predicts and writes each as a restriction on β:

- **Purchasing power parity**: $p_t - p_t^* - e_t \sim I(0)$ (domestic prices p, foreign prices p^*, exchange rate e, all in logs).
- **Uncovered interest parity**: $r_t - r_t^* \sim I(0)$.
- **Fisher equation**: $r_t - \Delta p_t \sim I(0)$ (stationary real rate).
- **Output gap**: domestic and foreign output share a trend.
- **Money demand**: $m_t - p_t - \beta_y y_t + \beta_r r_t \sim I(0)$.

With r cointegrating vectors, exact identification needs r^2 restrictions (r per vector, including one normalisation); restrictions beyond that are *over-identifying* and are tested by a likelihood-ratio test, asymptotically χ^2 with degrees of freedom equal to their number [147].

12.2 FROM THE LONG RUN TO THE SHORT RUN

Once β is fixed, the VECM's short-run part is estimated and can be given structure as in an SVAR. Because the long run is theory-consistent, impulse responses converge to economically sensible values, and the persistence profiles of each long-run relation show how fast it is restored after a shock [136].

12.3 WHY USE IT

Pure VARs have no long-run anchor; pure theory models impose dynamics the data may reject. The structural cointegrating VAR is a middle ground: interpretable equilibrium, flexible adjustment. It has been used for the UK economy, for forecasting at central banks and for global VAR (GVAR) models that link many country models through trade weights.

A SIMPLE EXAMPLE. Over a year Indian prices rise 6%, US prices 2%, and the rupee depreciates 4%. The real exchange rate $p - p^* - e$ changes by $6 - 2 - 4 = 0$: PPP holds. Imposing the cointegrating vector $(1, -1, -1)$ on (p, p^*, e) and testing it is one over-identifying restriction in a structural cointegrating VAR; if the likelihood-ratio test does not reject, the model's long run is PPP.

* * *

CHAPTER 13

FACTOR MODELLING APPROACH (FMA)

Large panels of N series $\mathbf{X}_t$ (hundreds of prices, production indices, survey balances) are summarised by a few common factors: $\mathbf{X}_t = \Lambda \mathbf{F}_t + \mathbf{e}_t$ [167]. **Analogy.** A stadium of people clapping: thousands of hands, but a few rhythms.

13.1 STATIC FACTOR MODELS

Factors load only contemporaneously. With r factors, the common component $\Lambda \mathbf{F}_t$ explains the co-movement and $\mathbf{e}_t$ is idiosyncratic. The number of factors is chosen by the Bai–Ng information criteria, e.g. $IC_{p2}(k) = \ln V(k) + k \frac{N+T}{NT} \ln \min(N,T)$ [5].

Application. Stock and Watson forecast US industrial production and inflation from factors extracted from about 215 monthly series ("diffusion indexes"), beating small VARs and leading-indicator models [167].

13.2 DYNAMIC FACTOR MODELS

Loadings are lag polynomials, $\mathbf{X}_t = \Lambda(L)\mathbf{f}_t + \mathbf{e}_t$, and factors follow their own VAR [51, 57, 159]. Nowcasting models at central banks use this form with a Kalman filter to update estimates of current GDP as each data release arrives.

13.3 NOWCASTING

Giannone, Reichlin and Small used a dynamic factor model with the Kalman filter to estimate current-quarter GDP from about 200 monthly indicators released at different dates, updating the estimate with each release (the "ragged edge") [235]. The approach underlies the nowcasts

published by several central banks.

A SIMPLE EXAMPLE. Three indicators load on one "economic activity" factor F: industrial production $X_1 = 1.0F + e_1$, retail sales $X_2 = 0.8F + e_2$, unemployment claims $X_3 = -0.5F + e_3$. If F rises by 2, the common component predicts $+2$, $+1.6$ and -1 respectively; the e's are series-specific noise. One number, the factor, summarises the co-movement of all three.

> IN PLAIN WORDS. Hundreds of series often move to the beat of a few hidden drummers. Factor models find the drummers.

* * *

Chapter 14

Methods of Extracting Factors

14.1 Principal component analysis (PCA)

Find the direction in the data with the largest variance, then the next orthogonal one, and so on: the eigenvectors of the sample covariance (or correlation) matrix [85, 137]. The first r principal components estimate the factor space consistently when $N, T \to \infty$ [167].

R example. For daily returns on the DAX, SMI, CAC and FTSE, the first principal component explains 74% of the (standardised) variance and loads almost equally on all four (0.48–0.52): a "European market" factor. The next three components explain only 11%, 9% and 6%.

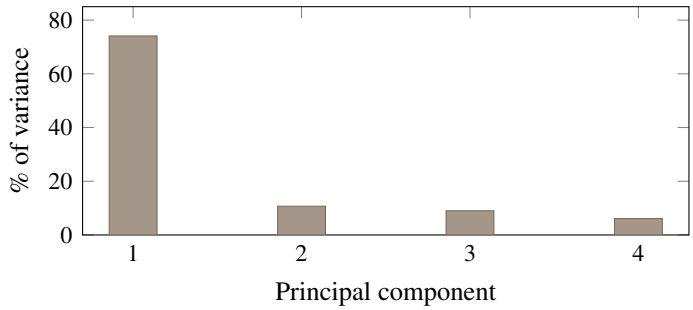


Figure 14.1: Scree plot for daily returns on the DAX, SMI, CAC and FTSE: the first principal component, a "European market" factor, explains 74% of the variance.

14.2 Bayesian likelihood methods

Treat the factors as unobserved states in a state-space model and estimate everything by Gibbs sampling: given the parameters, draw the factors with the Kalman filter and smoother; given the factors, draw the loadings and autoregressive coefficients by regression with priors; repeat. The output is a full posterior for the factors, so uncertainty about the factor is carried into any

forecast or index. Otrok and Whiteman built a Bayesian leading-indicator index for the Iowa economy this way [135]. The method also handles hierarchical structures (world, regional and country factors). The outline lists “Kohm and center (1994)”, probably Kim and Nelson, whose book treats state-space and Markov-switching factor models [98].

14.3 SPECTRAL ANALYSIS

Dynamic principal components work in the frequency domain. At each frequency ω, compute the eigenvectors of the estimated spectral density matrix of $\mathbf{X}_t$; the leading ones give filters that combine leads and lags of the series, so a factor can load on, say, retail sales immediately and on employment two months later. Forni, Hallin, Lippi and Reichlin’s generalised dynamic factor model is estimated this way [25, 51]. The cost is two-sided filters that use future data, which complicates real-time use; one-sided versions were developed for forecasting.

A SIMPLE EXAMPLE. Two standardised series with correlation 0.8 have correlation matrix $\left(\begin{smallmatrix}1 & 0.8\\ 0.8 & 1\end{smallmatrix}\right)$, with eigenvalues 1.8 and 0.2 and eigenvectors $(0.707, 0.707)$ and $(0.707, -0.707)$. The first principal component, the simple average of the two, explains $1.8/2 = 90\%$ of the total variance; the second, their difference, only 10%.

> IN PLAIN WORDS. PCA rotates the data until the biggest common movement lines up with the first axis.

* * *

CHAPTER 15

NONLINEAR TIME SERIES MODELS

15.1 NONLINEAR AUTOREGRESSIVE MODEL (NLAR)

$y_t = f(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$ with f nonlinear [173]. A nonlinear f can produce asymmetric cycles, limit cycles and regime-dependent persistence, none of which a linear AR can generate. Before fitting one, test linearity (Chapter 19).

15.2 GENERALISED NONLINEAR AUTOREGRESSIVE MODEL (GNLAR)

Allows the conditional variance, as well as the conditional mean, to depend nonlinearly on the past:

$$y_t = f(\mathbf{y}_{t-1}) + g(\mathbf{y}_{t-1})\,\varepsilon_t \qquad \varepsilon_t \sim \text{i.i.d.}(0,1),$$

so that the size of shocks varies with the state of the system [173]. ARCH effects (Chapter 20) are the special case where g^2 depends on past squared shocks. Such models describe series that are both cyclical and more volatile in some phases, such as commodity prices that are more volatile when stocks are low.

15.3 BILINEAR MODEL

$y_t = \sum_i \phi_i y_{t-i} + \sum_j \theta_j \varepsilon_{t-j} + \sum_{i,j} b_{ij} y_{t-i}\varepsilon_{t-j} + \varepsilon_t$ [62]. The cross terms make the effect of a shock depend on the current level of the series; bilinear series show occasional bursts of large values, similar to what is seen in some financial and geophysical data.

A SIMPLE EXAMPLE. Take the bilinear model $y_t = 0.5y_{t-1} + 0.5\,y_{t-1}\varepsilon_{t-1} + \varepsilon_t$. A shock $\varepsilon_{t-1} =$

1 adds nothing extra when $y_{t-1} = 0$, but adds $0.5 \times 2 \times 1 = 1$ when $y_{t-1} = 2$, doubling the linear effect of $0.5 \times 2 = 1$. The same shock matters more when the system is already far from zero.

IN PLAIN WORDS. In a nonlinear model, the same shock has different effects depending on where the system is when the shock hits.

* * *

CHAPTER 16

THRESHOLD MODELS

The dynamics switch between regimes according to whether a threshold variable z_{t-d} exceeds a threshold r:

$$y_t = \left(\phi_0^{(1)} + \sum_i \phi_i^{(1)} y_{t-i}\right) 1(z_{t-d} \le r) + \left(\phi_0^{(2)} + \sum_i \phi_i^{(2)} y_{t-i}\right) 1(z_{t-d} > r) + \varepsilon_t. \tag{16.1}$$

Analogy. A thermostat: below the set point the heater runs, above it the heater is off; the room's dynamics differ in the two regimes.

16.1 THRESHOLD REGRESSION MODEL

The regression version, with an exogenous threshold variable. Estimation by sample splitting: for each candidate r on a grid, fit OLS in both regimes, and choose the r that minimises the total sum of squared residuals. Hansen derived the distribution of $\hat{r}$ and likelihood-ratio confidence intervals [77]; in his growth application, countries split into regimes by initial income and literacy.

16.2 THRESHOLD AUTOREGRESSIVE (TAR) MODEL

Introduced by Tong for nonlinear time series [173, 174], motivated by cyclical data such as the Canadian lynx and sunspots.

16.3 SELF-EXCITING TAR (SETAR)

The threshold variable is a lag of the series itself, $z_{t-d} = y_{t-d}$. Chan proved that the least-squares estimate of r is super-consistent (converges at rate T) [29].

R EXAMPLE: the lynx. For $\log_{10}$ lynx trappings with two lags and delay $d = 2$, a grid search puts the threshold at $r = 3.31$ and cuts the residual sum of squares from 5.78 (linear AR(2)) to 4.35. In the lower regime ($y_{t-2} \le 3.31$, the population was small two years ago) the AR coefficients are $(1.26, -0.43)$, a persistent recovery; in the upper regime they are $(1.60, -1.01)$, a sharp oscillation that turns a peak into a crash: the slow rise and fast fall of the cycle.

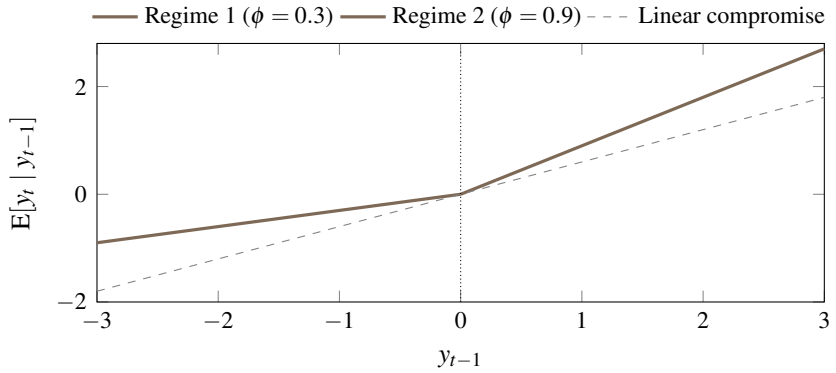


Figure 16.1: The threshold model of the simple example: tomorrow's expected value as a function of today's. The slope jumps at the threshold $r = 0$; a single straight line (dashed) fits neither regime.

16.4 AUTOREGRESSIVE SELF-EXCITING THRESHOLD MODEL

The general SETAR($k; p_1, \ldots, p_k$) has k regimes with their own lag orders p_j, intercepts and error variances, selected by the series' own lag y_{t-d}. Model building: choose the maximum lag; for each delay d test linearity (Tsay's arranged regression test); choose d with the strongest rejection; estimate thresholds by grid search and lag orders by AIC within each regime; check residuals [173]. Tong and Lim's lynx model is a SETAR(2;7,2) with delay 2 [174].

16.5 NON-SETAR (THRESHOLD VARIABLE NOT A LAG OF y)

TAR with an exogenous threshold variable, e.g. output dynamics that change when the interest rate or a credit spread crosses a level. Testing linearity is non-standard because r is not identified under the null; Hansen's sup-LM statistics with bootstrap p-values solve this [74, 75].

16.6 MULTIVARIATE SETAR

Extends the framework to systems: a VAR whose coefficients switch with a threshold variable (threshold VAR, TVAR) [177]. Used for asymmetric effects of monetary policy (tightening in a credit crunch versus normal times) and for nonlinear adjustment in cointegrated systems: in a threshold VECM, arbitrage between two markets operates only once the price gap exceeds transaction costs, so the error-correction coefficient is zero inside a band and negative outside it.

16.7 TSAY (1998), MULTIVARIATE TAR

Tsay generalises his arranged-regression test to vectors: order the observations by the threshold variable, compute recursive predictive residuals for the whole system, and test whether they are predictable from the regressors with a χ^2 statistic. The paper also gives a modelling procedure for selecting the delay, the threshold and the lag orders [177], and applies it to arbitrage between US stock index futures and the cash index.

16.8 MULTIPLE-REGIME TAR

More than two regimes, e.g. recession, normal growth and boom. Thresholds are estimated one at a time: fit a two-regime model, fix the first threshold, search for a second given the first, then re-estimate the first given the second (Bai's refinement); the number of regimes is chosen by sequential bootstrap tests [77, 176].

A SIMPLE EXAMPLE. Let $y_t = 0.9y_{t-1} + \varepsilon_t$ if $y_{t-1} > 0$ and $y_t = 0.3y_{t-1} + \varepsilon_t$ if $y_{t-1} \le 0$ (threshold $r = 0$, delay 1). Starting from $+2$ the path is $1.8, 1.62, 1.46$: positive deviations persist. Starting from -2 it is $-0.6, -0.18, -0.05$: negative deviations die out quickly. A linear AR fitted to such data would average the two speeds and describe neither.

> IN PLAIN WORDS. A threshold model has a switch: when a chosen variable crosses a line, the rules of motion change.

* * *

CHAPTER 17

SMOOTH TRANSITION MODELS

Replace the 0/1 indicator by a continuous transition function $G(s_t;\gamma,c)\in[0,1]$: $y_t=\phi'\mathbf{x}_t[1-G(s_t)]+\theta'\mathbf{x}_tG(s_t)+\varepsilon_t$, so regimes change gradually [64, 169]. The slope γ controls the speed: as $\gamma\to\infty$ the model becomes a threshold model, as $\gamma\to 0$ a linear one. Gradual change suits aggregate data, where millions of agents switch at slightly different points.

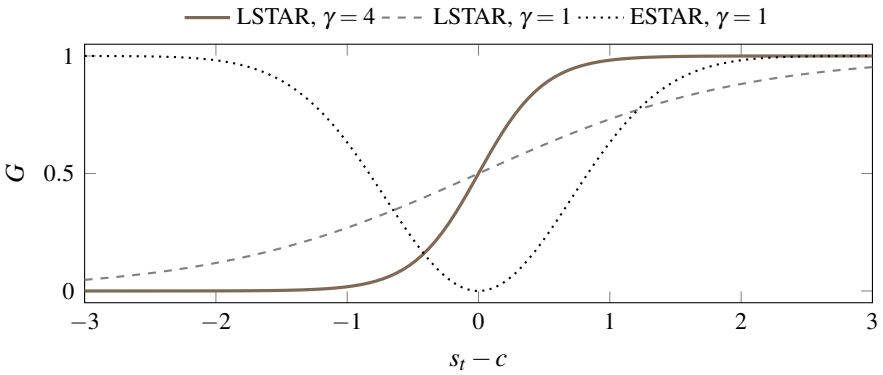


Figure 17.1: Logistic transition (asymmetric: one regime below c, another above) and exponential transition (symmetric: inner and outer regime).

17.1 SMOOTH TRANSITION REGRESSION MODEL (STR)

Granger and Teräsvirta's regression with smooth regime change [64]. Linearity is tested by a Taylor expansion of G around $\gamma=0$, which turns the problem into an auxiliary regression on products of $\mathbf{x}_t$ with powers of s_t [116].

17.2 STAR MODELS

Smooth transition autoregressions, with $s_t = y_{t-d}$; modelling cycle: specify a linear AR, test linearity, choose d and the transition function, estimate by nonlinear least squares, evaluate [179].

17.2.1 LOGISTIC STAR (LSTAR)

$G = [1+\exp\{-\gamma(s_t - c)\}]^{-1}$: different dynamics for low and high values of s_t, suitable for expansions versus recessions [169]. Teräsvirta and Anderson found LSTAR dynamics in industrial production for several OECD countries [170].

17.2.2 EXPONENTIAL STAR (ESTAR)

$G = 1-\exp\{-\gamma(s_t - c)^2\}$: symmetric, with an inner regime near c and an outer regime far from it [170]. The classic application is real exchange rates: near parity, transaction costs make arbitrage unprofitable and the rate wanders like a random walk; far from parity, arbitrage pulls it back fast. Taylor, Peel and Sarno showed that this nonlinear mean reversion can explain the slow average half-lives behind the PPP puzzle [195].

17.2.3 MULTIPLE-REGIME STAR (MRSTAR)

Several transition functions, each with its own transition variable or location, give more than two regimes, e.g. $y_t = \phi_1'\mathbf{x}_t + \phi_2'\mathbf{x}_t G_1(s_{1t}) + \phi_3'\mathbf{x}_t G_2(s_{2t}) + \varepsilon_t$ [179]. With two logistic functions in the same variable, the model can have a lower, middle and upper regime, as in a three-phase business cycle.

17.2.4 VECTOR STAR (VSTAR)

Multivariate extension of STAR in which the whole VAR coefficient matrix moves between regimes as a common (or equation-specific) transition function changes [179]. Used to model how the transmission of shocks between variables, such as interest rates and output, differs in recessions and expansions.

17.2.5 TIME-VARYING STAR (TVSTAR)

Adds a second transition in calendar time, $G_2(t/T)$, so that the STAR parameters also drift smoothly, for example because the business cycle became less volatile after the mid-1980s.

The model nests linear, STAR and smoothly time-varying linear models, each testable by the Taylor-expansion approach [115].

A SIMPLE EXAMPLE. Let the AR coefficient be $0.9(1-G)+0.3G$ with logistic G, $\gamma=2$, $c=0$. At $s=-1$: $G=1/(1+e^2)=0.12$ and the coefficient is 0.83. At $s=0$: $G=0.5$, coefficient 0.60. At $s=1$: $G=0.88$, coefficient 0.37. Persistence falls gradually as s rises, instead of jumping at a threshold.

> IN PLAIN WORDS. Smooth transition models are threshold models with a dimmer instead of a switch.

* * *

CHAPTER 18

REGIME SWITCHING MODELS (RSM)

The regime $s_t \in \{1,\dots,M\}$ is an *unobserved* Markov chain with transition matrix $\mathbf{P}$, $p_{ij} = \Pr(s_t = j \mid s_{t-1} = i)$. Parameters switch with the regime, and the probability of each regime at each date is inferred from the data by the Hamilton filter; estimation is by maximum likelihood or the EM algorithm [71]. The expected duration of regime i is $1/(1-p_{ii})$. Unlike threshold models, the switch is not tied to an observed variable crossing a line.

18.1 REGIME SWITCHING REGRESSION (RSR)

A regression whose coefficients and error variance depend on the regime, $y_t = \mathbf{x}_t'\beta_{s_t} + \sigma_{s_t}\varepsilon_t$. Quandt's original switching regression let the regime be independent over time; Goldfeld and Quandt made it a Markov chain, so regimes persist [60]. Applications include markets in and out of equilibrium (excess demand and excess supply regimes) and the relation between returns and risk factors in calm and turbulent markets.

18.2 MARKOV SWITCHING AR (MSAR)

$y_t - \mu(s_t) = \sum_i \phi_i\{y_{t-i} - \mu(s_{t-i})\} + \varepsilon_t$ [71]. Hamilton fitted this to quarterly US real GNP growth with two regimes: an expansion with mean growth of about 1.2% per quarter and a recession with about -0.4%. The estimated transition probabilities imply that expansions last about ten quarters and recessions about four, and the inferred recession probabilities line up closely with NBER recession dates.

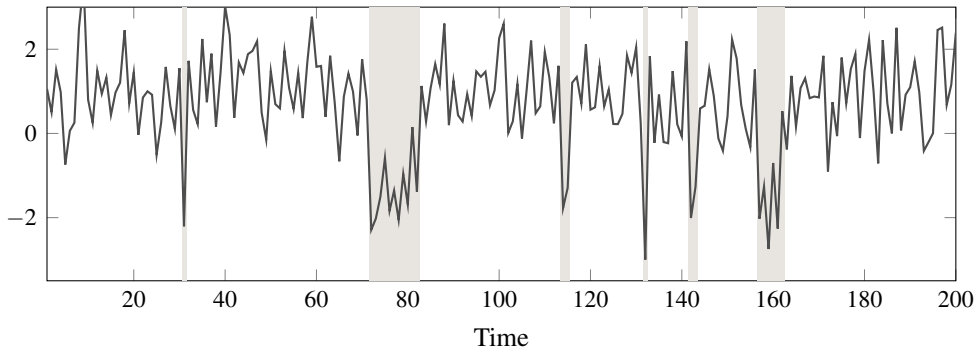


Figure 18.1: A simulated two-regime Markov-switching series: mean $+1$ in the normal regime and -1.5 in the shaded "recession" regime, with persistence $p_{11} = 0.95$, $p_{22} = 0.85$. The regime is never observed; the Hamilton filter infers it from the data.

18.3 MARKOV SWITCHING VAR (MS-VAR)

Multivariate regime switching in means, intercepts, coefficients or variances; Krolzig's notation, e.g. MSIAH(M)-VAR(p), lists what switches (I intercept, A autoregressive parameters, H heteroskedasticity) [102]. Used to date business cycles jointly across countries or sectors (a common regime driving several GDP series) and to model calm and crisis regimes in asset returns, where correlations as well as volatilities switch.

18.4 MARKOV SWITCHING VECM (MS-VECM)

Regime switching in an equilibrium-correction model: the long-run relation β is stable, but the intercept, the speed of adjustment α or the variance switch with the regime [103]. Krolzig, Marcellino and Mizon applied it to the UK labour market, where adjustment of employment, wages and productivity differed across business-cycle regimes.

18.5 TIME-VARYING TRANSITION PROBABILITIES

Let the transition probabilities depend on observed variables, $p_{ij,t} = \Lambda(\mathbf{z}'_{t-1}\delta_{ij})$, so that, for example, the chance of moving into recession rises as leading indicators weaken and expected durations change over time [236].

A SIMPLE EXAMPLE. With $p_{11} = 0.9$ (stay in expansion) and $p_{22} = 0.75$ (stay in recession), the expected durations are $1/(1-0.9) = 10$ quarters and $1/(1-0.75) = 4$ quarters, and in the long run the economy spends $0.1/(0.1+0.25) = 29\%$ of the time in recession. If a quarter of weak data raises the filtered recession probability from 0.2 to 0.7, the model has "called" a

recession without anyone announcing it.

> IN PLAIN WORDS. The economy has moods it does not announce. A Markov-switching model infers which mood it is in from how the data behave, and how likely the mood is to change.

* * *

Chapter 19

Artificial Neural Networks (ANN) and Tests for Nonlinearity

A single-hidden-layer network, $y_t = \beta_0 + \sum_{j=1}^{q} \beta_j g(\gamma_j' \mathbf{x}_t) + \varepsilon_t$, with logistic $g(u) = 1/(1+e^{-u})$, can approximate any continuous function on a compact set arbitrarily well as q grows (universal approximation) [84]. Weights are estimated by back-propagation, i.e. gradient descent on the squared error [156], with many random starts because the error surface has many local minima. The tests below check whether a linear model is adequate *before* nonlinear models are fitted: rejecting linearity is a licence to try a nonlinear model, not proof that a particular one is right.

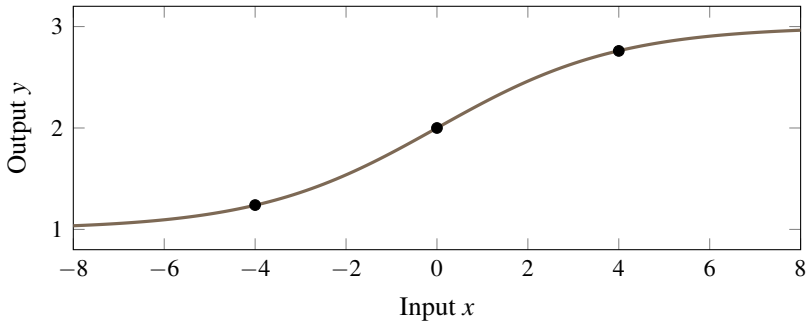


Figure 19.1: Output of the one-hidden-unit network of the simple example, $y = 1 + 2g(0.5x)$: an S-shaped curve that a linear model cannot reproduce. Dots mark the worked values.

19.1 Parametric tests

All follow one recipe: fit the linear AR, keep the residuals, regress them on candidate nonlinear terms, and F-test whether those terms help.

19.1.1 RAMSEY (1969) RESET TEST

Add powers of the fitted value, $\hat{y}_t^2, \ldots, \hat{y}_t^{s+1}$, to the regression and test their joint significance with s numerator degrees of freedom [154]. Keenan's one-degree-of-freedom version and Tsay's F test with all squares and cross-products of the lags [202] are its time-series descendants.

In the original paper. Ramsey [154] designed RESET to detect three specification errors: omitted variables, incorrect functional form, and correlation between regressors and the error. His original version used BLUS residuals (best linear unbiased residuals with a scalar covariance matrix) and regressed them on powers of the fitted values; Ramsey and Schmidt [265] showed that using ordinary least-squares residuals gives the same test, which is the F form used today.

19.1.2 TSAY (1989) TEST

Arranged-regression test for threshold nonlinearity: sort the observations by the threshold variable y_{t-d}, compute recursive one-step predictive residuals, and regress them on the regressors; a threshold makes them predictable [176].

In the original paper. Tsay [176] introduced the arranged autoregression so that a threshold model with unknown threshold becomes a regression with an unknown change point in the ordered data. Besides the F test, he proposed a modelling strategy: choose the delay d with the largest F, then locate the threshold from scatterplots of the recursive t-ratios of the AR coefficients against the threshold variable, where a threshold shows up as a change in the path.

19.1.3 THRESHOLD TEST

Sup-type LM, Wald or LR tests over the unidentified threshold, with bootstrap p-values [74, 176]. The neural-network linearity tests of Lee–White–Granger (random hidden units) and Teräsvirta–Lin–Granger (a Taylor expansion of the network) are also widely used [108, 171].

Application. On the lynx series, tests built on the lags (Tsay's F test, the threshold test) reject linearity decisively ($p \approx 0.0001$), while RESET is borderline ($p \approx 0.06$): the direction in which a test looks matters.

19.2 NON-PARAMETRIC TESTS

19.2.1 MCLEOD–LI (1983)

Ljung–Box statistic on squared residuals, $Q = n(n+2)\sum_{k=1}^{m} \hat{r}_k^2(\hat{\varepsilon}^2)/(n-k) \sim \chi_m^2$; detects ARCH-type (variance) nonlinearity [124].

In the original paper. McLeod and Li [124] showed that the autocorrelations of squared residuals from a fitted ARMA model are asymptotically independent normal with variance $1/n$, unaffected by the estimation of the ARMA parameters, so the portmanteau statistic is χ_m^2 with no degrees-of-freedom correction. They proposed it as a diagnostic for nonlinearity after an ARMA fit.

19.2.2 BISPECTRAL TEST

For a linear process with independent shocks the normalised bispectrum (bicoherence) is constant across frequency pairs, and zero if the process is Gaussian; Hinich's tests check both [82].

19.2.3 BDS STATISTIC

Tests whether residuals are i.i.d. using the correlation integral $C_m(\varepsilon)$, the share of pairs of m-histories within distance ε:

$$BDS_{m,\varepsilon} = \sqrt{T}\,[C_m(\varepsilon) - C_1(\varepsilon)^m]/\sigma_m \to N(0,1)$$

[24]. Under independence, two stretches of m days are close with probability C_1^m; hidden structure makes close stretches more common. Hsieh used it to show that stock returns are not i.i.d., mainly because of volatility clustering [204].

In the original paper. Brock, Dechert, Scheinkman and LeBaron [24] took the correlation integral from the analysis of chaotic systems and turned it into a statistical test: using the theory of U-statistics they proved that $\sqrt{T}[C_m(\varepsilon) - C_1(\varepsilon)^m]$ is asymptotically normal under i.i.d. data. The test has power against linear dependence, nonlinear dependence and deterministic chaos; applied to residuals of a fitted model, its distribution is unaffected by estimation for many models, though not for standardised GARCH residuals, which need adjusted critical values.

19.3 DEEP LEARNING AND MACHINE LEARNING

Recurrent networks with long short-term memory (LSTM) cells [237] learn long-range dependence that simple feed-forward AR-ANNs miss, and are now common in high-frequency and textual forecasting. In finance, Gu, Kelly and Xiu compared penalised regressions, trees, random forests and neural networks for predicting US stock returns; trees and neural networks performed best, in some cases doubling the economic gains of leading regression-based strategies, mainly by capturing nonlinear interactions between predictors [238]. The tests above remain useful as a first check that nonlinearity is there to be learned.

A SIMPLE EXAMPLE. A network with one hidden unit, $y = 1 + 2g(0.5x)$ and logistic g: at $x = 0$, $g(0) = 0.5$ and $y = 2$; at $x = 4$, $g(2) = 0.88$ and $y = 2.76$; at $x = -4$, $g(-2) = 0.12$ and $y = 1.24$. The output is an S-shaped curve in x, steep near zero and flat at the extremes, which no straight line can reproduce.

> IN PLAIN WORDS. Before building a curved model, check that the straight one is really bent. Each test looks for a different kind of bend.

* * *

Chapter 20

Volatility Models

20.1 Univariate volatility models

Model the conditional variance $\sigma_t^2 = \mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1})$. Financial returns show volatility clustering, fat tails and a leverage effect (falls raise volatility more than rises).

20.1.1 ARCH

$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2$ [42]. Engle's original application was UK inflation, whose uncertainty varied over time. His LM test regresses $\hat{\varepsilon}_t^2$ on q lags; $TR^2 \sim \chi_q^2$.

20.1.2 GARCH

$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta < 1$, long-run variance $\omega/(1-\alpha-\beta)$, half-life $\ln 0.5 / \ln(\alpha+\beta)$ [18]. GARCH(1,1) is an ARCH(∞) with geometrically declining weights. For daily DAX returns 1991–98, $\alpha = 0.069$, $\beta = 0.887$: a volatility shock has a half-life of about 15 trading days.

20.1.3 GJR (TGARCH)

$\sigma_t^2 = \omega + (\alpha + \gamma 1_{\varepsilon_{t-1}<0})\varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, where $\gamma > 0$ captures leverage [59, 189].

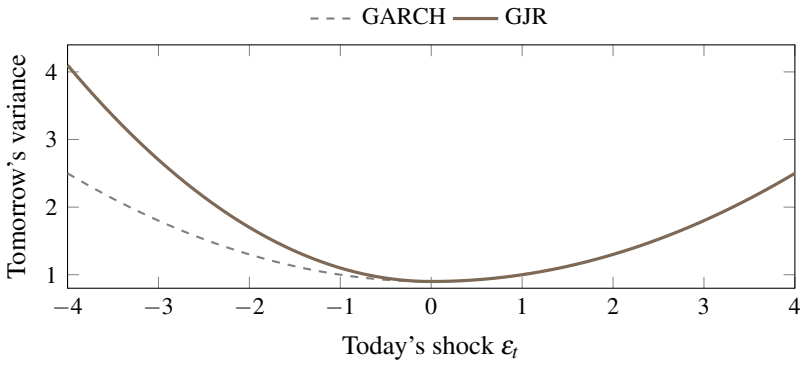


Figure 20.1: News impact curves for the simple example ($\omega = 0.1$, $\alpha = 0.1$, $\beta = 0.8$, $\theta = 0.1$, previous variance 1): GJR raises next-day variance more after bad news than after good news of the same size.

20.1.4 EGARCH

$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha(|z_{t-1}| - \mathrm{E}|z_{t-1}|) + \gamma z_{t-1}$; no positivity constraints, and $\gamma < 0$ means bad news raises volatility more [131].

20.1.5 APARCH

$\sigma_t^\delta = \omega + \sum \alpha_i (|\varepsilon_{t-i}| - \gamma_i \varepsilon_{t-i})^\delta + \sum \beta_j \sigma_{t-j}^\delta$, which nests ARCH, GARCH, GJR, TGARCH and several others by fixing δ, γ_i and β_j [38].

20.1.6 VALUE AT RISK (VAR)

The loss that will not be exceeded with confidence $1-\alpha$ over a horizon. With conditional mean μ_t, volatility σ_t and the α-quantile q_α of the standardised distribution ($q_{0.01} = -2.326$ for the normal), the return quantile is $\mu_t + \sigma_t q_\alpha$ and the VaR, stated as a positive loss, is $\mathrm{VaR}_t(\alpha) = -(\mu_t + \sigma_t q_\alpha)$ [95]. CAViaR models the quantile directly by regression quantiles [47]. The outline's "VAR" here means Value at Risk, not vector autoregression.

Example. A Rs 1 crore position with daily volatility 1.99% has a one-day 95% VaR of $1.645 \times 0.0199 \times 10^7 =$ Rs 3.27 lakh. Backtests on the DAX show that normal-GARCH VaR is breached too often at 99%, while GARCH with Student-t shocks passes: the variance model gets the scale right, the distribution gets the tail right.

20.2 MULTIVARIATE VOLATILITY MODELS

Models for the conditional covariance matrix $\mathbf{H}_t$, which must stay positive definite and have a manageable number of parameters.

20.2.1 VEC: VECH AND DIAGONAL VEC

$\text{vech}(\mathbf{H}_t) = \mathbf{C} + \mathbf{A}\,\text{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + \mathbf{B}\,\text{vech}(\mathbf{H}_{t-1})$ [20]: 21 parameters for two assets, 6,105 for ten. The diagonal version lets each element follow its own GARCH.

20.2.2 BEKK

$\mathbf{H}_t = \mathbf{C}'\mathbf{C} + \mathbf{A}'\varepsilon_{t-1}\varepsilon_{t-1}'\mathbf{A} + \mathbf{B}'\mathbf{H}_{t-1}\mathbf{B}$; positive definite by construction, and off-diagonal elements of $\mathbf{A}$ measure volatility spillovers [46].

20.2.3 CONSTANT CONDITIONAL CORRELATION (CCC)

$\mathbf{H}_t = \mathbf{D}_t\mathbf{R}\mathbf{D}_t$ with univariate GARCH volatilities in $\mathbf{D}_t$ and $\mathbf{R}$ constant [19].

20.2.4 DYNAMIC CONDITIONAL CORRELATION (DCC)

$\mathbf{Q}_t = (1-a-b)\bar{\mathbf{Q}} + a\mathbf{z}_{t-1}\mathbf{z}_{t-1}' + b\mathbf{Q}_{t-1}$, $\mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2}\mathbf{Q}_t\,\text{diag}(\mathbf{Q}_t)^{-1/2}$ [43]. For DAX and FTSE the DCC correlation moves between 0.32 and 0.75 and peaks in turbulent periods, when diversification is needed most.

20.2.5 BLOCK DCC AND REGIME-SWITCHING DCC (RS-DCC)

Correlations with block structure (groups of assets share dynamics) [188], or driven by a Markov chain [141].

20.2.6 STOCHASTIC VOLATILITY MODEL

Volatility follows its own latent process, $h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta\eta_t$ with $r_t = e^{h_t/2}\varepsilon_t$ [168, 186]; estimated by Kalman-filter QML or MCMC.

20.2.7 FACTOR ARCH

Covariances driven by a few factors with ARCH dynamics, so that N assets need only $k \ll N$ volatility processes [48].

20.3 REALISED VOLATILITY AND THE HAR MODEL

With intraday data, the sum of squared five-minute returns gives a nearly error-free measure of each day's variance, the *realised variance* [239]. Volatility can then be modelled as an observed series. Corsi's heterogeneous autoregressive (HAR) model regresses tomorrow's realised volatility on its daily, weekly and monthly averages, $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(5)} + \beta_m RV_t^{(22)} + \varepsilon_{t+1}$, reflecting traders with different horizons; this simple regression mimics long memory and forecasts very well [240].

20.4 REALISED GARCH AND GARCH-MIDAS

Realised GARCH adds a realised measure to the GARCH equation and models it jointly with returns [241]. GARCH-MIDAS splits volatility into a short-run GARCH component and a long-run component driven by low-frequency macroeconomic data (inflation, industrial production) through mixed-data sampling [242], linking market risk to the macroeconomy.

A SIMPLE EXAMPLE. GARCH(1,1) with $\omega = 0.1$, $\alpha = 0.1$, $\beta = 0.8$: long-run variance $0.1/(1-0.9) = 1$ (daily volatility 1%). After a -3% day with $h_{t-1} = 1$, $h_t = 0.1 + 0.1 \times 9 + 0.8 \times 1 = 1.8$, volatility 1.34%. With a GJR asymmetry term $\theta = 0.1$ the same fall adds another $0.1 \times 9 = 0.9$: $h_t = 2.7$, volatility 1.64%. A $+3\%$ day would leave it at 1.34%. On a Rs 1 crore position the one-day 99% VaR moves from Rs 2.33 lakh ($2.326 \times 1\%$) to Rs 3.82 lakh ($2.326 \times 1.64\%$).

> IN PLAIN WORDS. Risk is not constant: it rises after shocks and fades slowly. Volatility models forecast that weather, and multivariate ones forecast how closely markets move together.

* * *

CHAPTER 21

PANEL DATA ANALYSIS

Panels follow N units (firms, countries, households) over T periods. They control for unobserved, time-invariant differences between units and give more variation to estimate from.

21.1 LINEAR PANEL MODEL

$y_{it} = \alpha_i + \mathbf{x}_{it}'\beta + \varepsilon_{it}$, $i = 1,\dots,N$, $t = 1,\dots,T$ [7, 87]. α_i captures everything about unit i that does not change over time: management quality, geography, culture.

21.2 FIXED AND RANDOM EFFECTS

Fixed effects (FE) removes α_i by the within transformation $y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$; it allows α_i to be correlated with $\mathbf{x}_{it}$ but cannot estimate effects of time-invariant regressors. **Random effects** (RE) treats α_i as a random draw uncorrelated with $\mathbf{x}_{it}$ and uses GLS, i.e. quasi-demeaning by $\theta = 1 - \sqrt{\sigma_\varepsilon^2/(\sigma_\varepsilon^2 + T\sigma_\alpha^2)}$; it is more efficient if that assumption holds, inconsistent if not [127].

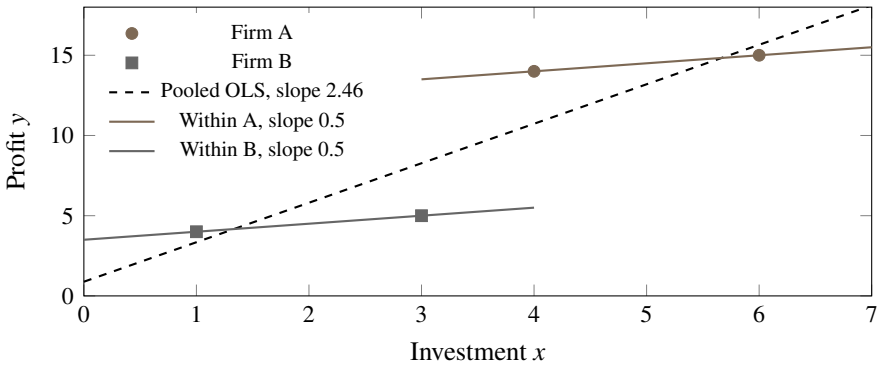


Figure 21.1: Why fixed effects matter (simple example). Pooling the two firms gives a steep slope because the better-run firm has both more investment and more profit; within each firm the slope is only 0.5.

21.3 HAUSMAN TEST (1978)

$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})'[\hat{\mathbf{V}}_{FE} - \hat{\mathbf{V}}_{RE}]^{-1}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) \sim \chi^2_k$: if the two estimates differ by more than sampling error, the RE assumption fails and FE is preferred [81].

R EXAMPLE: simulated firms. With 200 firms over 6 years, a true slope of 0.5, and firm effects correlated with the regressor (as when better-managed firms both invest more and earn more): pooled OLS gives 0.92 (badly biased), FE 0.46, RE 0.55, and the Hausman statistic is 69.1 ($p < 10^{-15}$), correctly pointing to FE.

Application. Cross-country growth regressions (e.g. the augmented Solow model of Mankiw, Romer and Weil [203]) were later re-estimated with country fixed effects, which typically changes the estimated speed of convergence considerably, because unobserved country characteristics are correlated with saving and schooling.

In the original paper. Hausman [81] gave a general principle: compare an estimator that is consistent under both the null and the alternative with one that is efficient under the null but inconsistent under the alternative. His key lemma is that the efficient estimator is uncorrelated with the difference between the two, so $\mathrm{Var}(\hat{\beta}_1 - \hat{\beta}_0) = \mathbf{V}_1 - \mathbf{V}_0$, with no covariance term to estimate. He applied the idea to errors in variables, simultaneous equations and, most famously, to fixed versus random effects in a panel wage equation.

21.4 DYNAMIC PANEL MODELS

With a lagged dependent variable, $y_{it} = \rho y_{i,t-1} + \mathbf{x}_{it}'\beta + \alpha_i + \varepsilon_{it}$, the within estimator is biased for small T (Nickell bias, of order $1/T$) [134]. Difference GMM first-differences away α_i and instruments $\Delta y_{i,t-1}$ with levels dated $t-2$ and earlier [3]; system GMM adds the level equa-

tion instrumented with lagged differences, which helps when ρ is close to one [4, 17]. Check with the Sargan/Hansen test of over-identifying restrictions and the AR(2) test on differenced residuals; too many instruments weaken both. Arellano and Bond's original application was employment equations for 140 UK companies.

21.5 PANEL ARDL

ARDL models for panels with large N and T, allowing heterogeneous short-run dynamics and error correction towards a long-run relation [149].

21.6 POOLED MEAN GROUP (PMG)

Long-run coefficients are restricted equal across units, while short-run coefficients, adjustment speeds and error variances vary [149]; the mean group (MG) estimator averages fully heterogeneous unit estimates [150], and a Hausman test chooses between them. Pesaran, Shin and Smith applied PMG to consumption functions in OECD countries and energy demand in Asian economies: the long run plausibly common, the short run country-specific.

21.7 CROSS-SECTIONAL DEPENDENCE: CCE AND INTERACTIVE FIXED EFFECTS

Common shocks (oil prices, global financial cycles) that affect units differently violate the independence behind standard panel estimators. Pesaran's common correlated effects (CCE) estimator adds cross-section averages of y and $\mathbf{x}$ as extra regressors to proxy the unobserved factors [243]; Bai's interactive fixed effects model, $y_{it} = \mathbf{x}_{it}'\beta + \lambda_i'\mathbf{F}_t + \varepsilon_{it}$, estimates the factors and loadings directly [244].

21.8 INSTRUMENT PROLIFERATION IN GMM

Difference and system GMM generate instruments that grow quadratically in T. Too many instruments overfit the endogenous variables, bias the estimates towards OLS/within and make the Hansen test implausibly perfect ($p \approx 1$). Roodman recommends collapsing the instrument matrix and limiting the lag depth, and reporting the instrument count [245].

21.9 DIFFERENCE-IN-DIFFERENCES WITH STAGGERED TREATMENT

When units are treated at different dates, the standard two-way fixed effects regression compares newly treated units with already-treated ones, and with heterogeneous effects it can even get the sign wrong. Goodman-Bacon decomposes the estimator into all its 2×2 comparisons [247]; Callaway and Sant'Anna estimate group–time average treatment effects using only clean comparisons and aggregate them [246]. These methods are now standard for evaluating staggered policy changes such as state-level regulations.

A SIMPLE EXAMPLE. Two firms, two years each. Firm A (well managed): investment $x = 4, 6$, profit $y = 14, 15$. Firm B: $x = 1, 3$, $y = 4, 5$. Pooled OLS on the four points gives a slope of $32/13 = 2.46$, because the well-run firm has both more investment and more profit. Within each firm, investment rises by 2 and profit by 1: the fixed-effects slope is 0.5. The pooled estimate confuses "good firms invest more" with "investment raises profit"; fixed effects compare each firm only with itself.

> IN PLAIN WORDS. Panels let each unit act as its own control. Fixed effects compare each firm with itself over time, which removes everything constant about the firm, observed or not.

* * *

Chapter 22

Panel Unit Root Tests without Breaks

Panel tests gain power by pooling cross-sections: 20 countries $\times$ 30 years carry more information about mean reversion than 30 years of one country. They differ in whether the autoregressive root is common ($\rho_i = \rho$) or heterogeneous, and in whether units are assumed independent (first generation) or allowed to be correlated (second generation). A key application is purchasing power parity: single-country tests rarely reject a unit root in real exchange rates, panel tests often do.

22.1 Levin and Lin (1992)

The working-paper predecessor of the Levin–Lin–Chu test (the outline writes "Levin and Lee") [109]. It established the pooled ADF approach with a common autoregressive root and a bias adjustment of the pooled t-statistic.

22.2 Levin–Lin–Chu

Common root: $\Delta y_{it} = \delta y_{i,t-1} + \text{unit-specific deterministics and lags} + \varepsilon_{it}$, H_0: $\delta = 0$ for all i against H_1: $\delta < 0$ for all i [110]. The alternative is restrictive: every unit reverts at the same speed.

In the original paper. Levin, Lin and Chu [110] allow unit-specific intercepts, trends and serial correlation, but a common autoregressive root. Their three steps: run ADF regressions unit by unit and keep orthogonalised residuals; normalise them by each unit's residual standard deviation; pool and compute the t-statistic on the common root, adjusted by a mean and standard deviation they tabulate. The asymptotics require $\sqrt{N}/T \to 0$, and they recommend the test for

panels of moderate size (roughly 10–250 units with 25–250 periods).

22.3 IM–PESARAN–SHIN

Heterogeneous roots: average the individual ADF t-statistics, $\bar{t} = N^{-1}\sum t_i$, standardised to $N(0,1)$ [88]. H_1: at least some units are stationary.

In the original paper. Im, Pesaran and Shin [88] replaced the common root by heterogeneous roots under the alternative, with a non-zero fraction of units stationary. The $\bar{t}$ statistic is standardised with the mean and variance of the individual ADF statistics, which they computed by simulation for each T and lag order, so the test has good size even for small T. Cross-sectional independence is assumed; they suggest subtracting cross-section means to absorb common time effects.

22.4 MADDALA–WU

Fisher-type test combining individual p-values, $P = -2\sum_{i=1}^{N} \ln p_i \sim \chi^2_{2N}$ [120]; works with unbalanced panels and any unit-root test.

In the original paper. Maddala and Wu [120] revived Fisher's 1932 method of combining p-values. They stressed its practical advantages over IPS (any unit-root test can be used, lag lengths and sample periods can differ by unit, the panel can be unbalanced), found by simulation that it often has better size and power, and proposed bootstrap critical values when units are cross-correlated.

22.5 HARRIS–TZAVALIS

Most panel unit-root tests assume both N and T large. Micro panels have many firms or households but few periods. Harris and Tzavalis derive the distribution of the pooled least-squares estimator of a common root for fixed T and large N, correcting the within-group (Nickell) bias analytically [78]. Use it for short panels, say $T < 25$.

22.6 BREITUNG

LLC needs a bias correction when the model includes individual trends, and that correction costs power. Breitung transforms the data (forward orthogonalisation of the differences and a detrending that does not use future information) so that no correction is needed; the resulting

test has higher local power against stationary alternatives with trends [22].

22.7 HADRI

Reverses the null, as KPSS does for one series: H_0 is stationarity in *all* units, H_1 a unit root in some. The statistic averages the unit-specific KPSS statistics and standardises them [70]. It is the confirmatory counterpart of IPS, but it over-rejects badly when errors are serially correlated or T is small, so treat rejections with care.

In the original paper. Hadri [70] generalised the KPSS statistic to panels: each unit's LM statistic is computed from the residuals of a regression on deterministic terms, the statistics are averaged, and the average is standardised with the moments of the KPSS limiting distribution. Versions allow error variances to differ across units.

22.8 CHOI

Combines the p-values of unit-by-unit tests with three statistics: Fisher's $P = -2\sum \ln p_i$, the inverse-normal $Z = N^{-1/2}\sum \Phi^{-1}(p_i)$, and a logit statistic [31]. The inverse-normal Z is recommended: it is well sized and powerful for both small and large N, and the panel can be unbalanced.

22.9 SECOND-GENERATION TESTS: CIPS

First-generation tests assume independent units, which fails for real exchange rates or GDP across countries. Pesaran's CIPS test augments each unit's ADF regression with cross-section averages (the CADF regression) and averages the t-statistics [248]. Test for dependence first with Pesaran's CD statistic [249].

A SIMPLE EXAMPLE. Ten countries' real exchange rates are tested one by one. Five give ADF p-values of 0.10 and five of 0.30: none rejects at 5%. Maddala–Wu combine them: $P = -2[5\ln 0.10 + 5\ln 0.30] = -2(-11.51 - 6.02) = 35.1$, above the 5% critical value of χ^2_{20}, 31.4. Together the countries reject the unit root even though no single one does: evidence that at least some real exchange rates revert to PPP.

> IN PLAIN WORDS. One short series rarely has enough information to tell slow mean reversion from a random walk; many short series together often do. Be clear whether you are claiming that all, some or none of the units are stationary.

Chapter 23

Panel Cointegration Tests without Breaks

23.1 McCoskey–Kao

An LM test with the null of cointegration in all units, the panel analogue of Shin's residual-based test (Chapter 5). The cointegrating regression is estimated efficiently (fully modified OLS or dynamic OLS) and the KPSS-type statistic is averaged across units [123].

23.2 Kao

Dickey–Fuller and ADF-type tests on the pooled residuals of a panel regression with individual intercepts and a *homogeneous* slope β [97]. Simple, but the assumption of one common cointegrating vector for all units is strong: if slopes differ, the pooled residual can look non-stationary even when each unit is cointegrated.

In the original paper. Kao [97] first showed that spurious regression also arises in panels: with independent $I(1)$ series, the within (LSDV) estimator converges to a constant rather than diverging, but its t-statistic diverges, so "significant" relations appear where none exist. He then proposed DF- and ADF-type tests on the pooled residuals under a homogeneous cointegrating vector.

23.3 Pedroni

Seven residual-based statistics, four pooling within the dimension (panel v, ρ, PP, ADF) and three between the dimension (group ρ, PP, ADF), allowing heterogeneous intercepts, trends

and slopes [138, 139]; Pedroni's own application was PPP.

In the original paper. Pedroni [138, 139] allowed the cointegrating slopes, intercepts, trends and dynamics to differ across units, so the null of no cointegration does not impose a common relation. The *within-dimension* (panel) statistics pool the autoregressive coefficients of the residuals and so assume a common root under the alternative; the *between-dimension* (group-mean) statistics average unit-by-unit statistics and allow heterogeneous roots. Each statistic, standardised with simulated moments, is asymptotically normal.

23.4 LARSSON–LYHAGEN–LÖTHGREN

The panel version of Johansen's trace test. Run the Johansen test in each unit, average the trace statistics, and standardise by the mean and variance of the asymptotic trace distribution: $\Upsilon_{LR} = \sqrt{N}(\overline{LR} - \mathrm{E}[Z])/\sqrt{\mathrm{Var}(Z)} \to N(0,1)$ [105]. It allows several cointegrating vectors and heterogeneous long-run relations, but needs reasonably long T for each unit.

23.5 BANERJEE–MARCELLINO–OSBAT

Show that cross-unit cointegration (e.g. between countries' price levels) badly distorts the size of panel unit root and cointegration tests that assume independence [9].

23.6 BAI AND NG (2004)

PANIC: separate common factors and idiosyncratic components by principal components and test each for unit roots, so non-stationarity can be traced to the common or the unit-specific part [6].

23.7 WESTERLUND (2005)

Variance-ratio tests that compare the variance of partial sums of residuals with their overall variance. They need no estimate of the long-run variance and no correction for serial correlation, and come in panel (pooled) and group-mean versions [180].

23.8 WESTERLUND (2007)

Four error-correction-based tests (G_t, G_a, P_t, P_a): is the adjustment coefficient zero? Bootstrap critical values handle cross-section dependence [182].

In the original paper. Westerlund [182] explained why residual-based tests can have low power: they impose a "common factor restriction" (the short-run dynamics of the levels regression must equal those of the regressors), which often fails. Testing directly whether the error-correction coefficient is zero avoids it. The group-mean statistics G_t, G_a test against cointegration in at least one unit; the panel statistics P_t, P_a against cointegration in the panel as a whole; bootstrap critical values handle cross-sectional dependence.

A SIMPLE EXAMPLE. Five countries' error-correction coefficients are -0.20, -0.30, -0.10, -0.25 and -0.15. Their average, -0.20, means 20% of any disequilibrium is closed each year, a half-life of $\ln 0.5 / \ln 0.8 = 3.1$ years. Westerlund's group-mean test asks whether this average is significantly below zero; if it is, the panel is cointegrated on average.

> IN PLAIN WORDS. Panel cointegration asks whether the same long-run relation holds across many units, and must be careful not to mistake common global shocks for evidence.

* * *

Chapter 24

Panel Cointegration Tests with Breaks

24.1 Carrion-i-Silvestre, Barrio-Castro and López-Bazo

A panel stationarity test (a Hadri-type null of stationarity) that allows up to five breaks in the level and/or trend of *each* unit, at dates that can differ across units. Break dates and numbers are estimated unit by unit with the Bai–Perron procedure, and bootstrap critical values handle cross-section dependence [28]. Applied to GDP per capita, it shows that allowing breaks overturns many apparent unit roots.

24.2 Banerjee and Carrion-i-Silvestre (2006)

Panel cointegration tests that allow structural breaks in the cointegrating relation and cross-sectional dependence through common factors [8]. The journal version appeared in 2015 (item 24.6 below).

24.3 Westerlund and Edgerton (2006)

LM-based tests of the null of no cointegration in a single equation, allowing for a structural break in the level and in the cointegrating slope [183]. They have good size and power and are often used as the time-series building block for panel tests.

24.4 WESTERLUND (2006)

Panel cointegration with multiple structural breaks: an LM test of the null of cointegration that allows breaks in the level and trend of each unit at unknown, unit-specific dates, estimated by minimising the sum of squared residuals [181]. It is the panel counterpart of testing for cointegration with breaks unit by unit, but with much more power.

24.5 BAI AND CARRION-I-SILVESTRE (2009)

Multiple structural breaks in each unit combined with common factors: the panel is split into common and idiosyncratic components, the breaks are allowed in the deterministic terms, and unit roots are tested in each component [221].

24.6 BANERJEE AND CARRION-I-SILVESTRE (2015)

Panel cointegration tests allowing structural breaks in the cointegrating relation and cross-section dependence through common factors [222]; implemented in the R package `xtbreakcoint`.

A SIMPLE EXAMPLE. Three countries' GDP-per-capita series have trend breaks in 1991 (India, liberalisation), 2008 (the UK, financial crisis) and 2016 (Brazil, recession). A panel test that forces one common break date, or none, will misread at least two of them as unit roots. Allowing unit-specific breaks lets each country's history speak for itself.

> IN PLAIN WORDS. Long panels span crises and policy changes; allowing breaks keeps a shifting relation from being mistaken for no relation.

* * *

CHAPTER 25

PANEL VECTOR AUTOREGRESSIVE MODEL (PVAR)

Idea. A panel VAR (PVAR) is a VAR estimated on many units (firms, banks, countries) at once. It keeps the VAR's "every variable explains every variable" structure and adds the panel's ability to control for unobserved differences between units. The price is an assumption: the dynamics ($\mathbf{A}_j$) are the same for every unit, while the levels (α_i) may differ [1, 83, 113].

25.1 THE MODEL

For unit i and period t, with a $K \times 1$ vector $\mathbf{y}_{it}$,

$$\mathbf{y}_{it} = \alpha_i + \mathbf{A}_1 \mathbf{y}_{i,t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{i,t-p} + \mathbf{B}\mathbf{x}_{it} + \varepsilon_{it}, \qquad \mathrm{E}[\varepsilon_{it}\varepsilon_{it}'] = \Sigma. \tag{25.1}$$

Holtz-Eakin, Newey and Rosen introduced the model to study the dynamic relation between local government revenues and expenditures [83].

25.2 WHY OLS FAILS: THE FIXED-EFFECT PROBLEM

The fixed effect α_i is correlated with the lagged dependent variables (a unit with a high α_i has high $\mathbf{y}$ in every period). Removing it by subtracting unit means creates the Nickell bias of order $1/T$ (Chapter 21), which is serious in short panels. First differences remove α_i but magnify gaps in unbalanced panels.

25.3 FORWARD ORTHOGONAL DEVIATIONS (HELMERT TRANSFORMATION)

Subtract the mean of all *future* observations and rescale so the errors stay homoskedastic:

$$\tilde{y}_{it} = c_t\Big(y_{it} - \frac{1}{T_i - t}\sum_{s>t} y_{is}\Big), \qquad c_t = \sqrt{\frac{T_i - t}{T_i - t + 1}}. \tag{25.2}$$

The fixed effect cancels, and since the transformed variable uses only future values, lagged *levels* ($\mathbf{y}_{i,t-1}$, $\mathbf{y}_{i,t-2}$, and so on) remain valid instruments. The model is then estimated equation by equation, or as a system, by GMM [1, 113].

25.4 SPECIFICATION AND INFERENCE

- **Lag order**: chosen by moment-and-model selection criteria (MMSC), the GMM analogues of AIC/BIC [263], and the Hansen J test of over-identifying restrictions.
- **Stability**: all eigenvalues of the companion matrix inside the unit circle.
- **Granger causality**: Wald tests on blocks of $\mathbf{A}_j$, now pooled across units.
- **Impulse responses and FEVD**: as in Chapter 8, with Cholesky or other identification; confidence bands by Monte Carlo draws from the GMM estimates.

25.5 APPLICATIONS

Love and Zicchino estimated a PVAR in firms' investment, sales, cash flow and Tobin's q for firms in 36 countries and found that investment responds more strongly to internal cash flow in countries with less developed financial systems, a sign of financing constraints [113]. Other common uses are bank lending and capital, house prices and credit across regions, and macro-financial linkages across countries.

25.6 LIMITS

The homogeneity assumption can be strong: if dynamics differ across units, a pooled PVAR averages them in a way that can be biased (the heterogeneity bias of [150]). Mean-group PVARs, Bayesian PVARs with partial pooling [26], and structural panel VARs (Chapter 27) relax it.

A SIMPLE EXAMPLE. The Helmert transformation in numbers. A firm's investment over four

years is $2,4,6,8$. For $t=1$: $\sqrt{3/4}\,(2-\frac{4+6+8}{3}) = 0.866\times(-4) = -3.46$; $t=2$: $\sqrt{2/3}\,(4-7) = -2.45$; $t=3$: $\sqrt{1/2}\,(6-8) = -1.41$. Add a fixed effect of 10 to every year $(12,14,16,18)$ and the transformed values are identical: the fixed effect is gone, and $y_{i,t-1}$ in levels is still a valid instrument because the transformation uses only later years.

* * *

CHAPTER 26

PANEL VECTOR ERROR CORRECTION MODEL (PVECM)

Idea. When the variables of a panel are $I(1)$ and cointegrated within each unit, a panel VECM keeps the long-run equilibrium and lets each unit adjust towards it [14, 105]. It is the panel counterpart of Chapter 9, and the natural model after a panel cointegration test (Chapter 23) finds cointegration.

26.1 THE MODEL

$$\Delta \mathbf{y}_{it} = \mu_i + \alpha_i \beta_i' \mathbf{y}_{i,t-1} + \sum_{j=1}^{p-1} \Gamma_{ij} \Delta \mathbf{y}_{i,t-j} + \varepsilon_{it}. \tag{26.1}$$

Three choices define the specification:

- **Long run**: common β (same equilibrium for all units, as theory often suggests, e.g. a PPP or money-demand relation) or unit-specific β_i;
- **Adjustment**: unit-specific α_i (different speeds of correction) is the usual choice;
- **Short run**: common or unit-specific Γ_{ij}.

A common β with heterogeneous α_i and Γ_{ij} is exactly the pooled mean group idea of Chapter 21 [149].

26.2 ESTIMATION

- **Rank**: the panel trace test of Larsson, Lyhagen and Löthgren averages unit-specific Johansen statistics [105].

- **Long-run vectors**: panel fully modified OLS [260] and panel dynamic OLS [261] estimate a common β with corrections for endogeneity and serial correlation; Breitung's two-step estimator estimates the VECM directly [262].
- **Short panels**: Binder, Hsiao and Pesaran give a quasi-maximum-likelihood estimator for panel VARs with unit roots and cointegration when T is small [14].

26.3 APPLICATIONS

Energy consumption and output across countries (is the long-run elasticity common?); money demand in the euro area; health spending and income across OECD countries; exchange rates and relative prices (panel PPP). Weak exogeneity tests on α_i show which variable does the adjusting, and so the direction of long-run causality.

A SIMPLE EXAMPLE. Suppose energy use and GDP share a common long-run elasticity of 0.8, but adjust at different speeds: $\alpha = -0.1$ in country A, -0.4 in country B. After GDP rises by 10%, energy use should eventually be 8% higher in both. Country B closes 40% of the gap each year (half-life 1.4 years), country A only 10% (half-life 6.6 years). A panel VECM with common β and unit-specific α_i captures exactly this.

* * *

CHAPTER 27

STRUCTURAL PANEL VAR

Idea. A structural panel VAR identifies named shocks (monetary, demand, supply) in many units at once and asks how their effects differ across units, rather than assuming they are the same [26, 140].

27.1 WHY A SEPARATE METHOD

A pooled PVAR with Cholesky identification gives one average response. But a monetary tightening may hit a highly indebted country much harder than a low-debt one, and policymakers want the whole distribution of responses. Shocks may also be partly common (a global oil shock) and partly local.

27.2 PEDRONI'S APPROACH

1. Estimate a reduced-form VAR for each unit separately, and one for the cross-section averages $\bar{\mathbf{y}}_t$.
2. Identify structural shocks in each unit and in the average, using any SVAR scheme (short-run, long-run or sign restrictions).
3. Decompose each unit's structural shocks into a *common* part, driven by the shocks of the cross-section average, and an *idiosyncratic* part: $\mathbf{u}_{it} = \Lambda_i \bar{\mathbf{u}}_t + \tilde{\mathbf{u}}_{it}$, with the loadings Λ_i estimated unit by unit.
4. Report the responses of each unit to common and to idiosyncratic shocks, and summarise them by the median and quantiles across units [140].

Because every unit has its own dynamics, the method avoids the heterogeneity bias of pooled

estimators [150], and needs moderately long time series for each unit.

27.3 ALTERNATIVES

Mean-group SVARs (estimate unit by unit, average the responses); Bayesian hierarchical panel VARs that shrink unit-specific coefficients towards a common mean [26]; and pooled SVARs when T is too short for unit-by-unit estimation.

27.4 APPLICATIONS

The transmission of monetary policy across euro-area countries; the effects of oil supply shocks on oil importers and exporters; and the response of regional house prices to national credit shocks. In each case the cross-unit spread of responses is the main finding, not the average.

A SIMPLE EXAMPLE. A 25-basis-point monetary tightening is identified in each of 12 countries. The median response of output after one year is -0.3%; the 10th and 90th percentiles across countries are -0.6% and -0.1%. The structural panel VAR reports all three: the average effect, and how much more the most indebted economies are hurt than the least.

* * *

CHAPTER 28

FACTOR-AUGMENTED PANEL VAR (FA-P-VAR)

Idea. Small VARs omit information that policy-makers and markets use, which distorts identified shocks. A factor-augmented VAR (FAVAR) summarises a large data set by a few principal-component factors and adds them to the VAR; the panel version does the same for a panel VAR, using factors to absorb the common shocks that create cross-sectional dependence [26, 187].

28.1 THE FAVAR MODEL

Let $\mathbf{X}_t$ be N informational series (say 120 prices, activity and financial indicators), driven by k unobserved factors $\mathbf{F}_t$ and the observed policy variable R_t:

$$\mathbf{X}_t = \Lambda^f \mathbf{F}_t + \Lambda^r R_t + \mathbf{e}_t, \qquad \begin{pmatrix} \mathbf{F}_t \\ R_t \end{pmatrix} = \Phi(L) \begin{pmatrix} \mathbf{F}_{t-1} \\ R_{t-1} \end{pmatrix} + \mathbf{v}_t. \tag{28.1}$$

28.2 ESTIMATION

1. **Two-step**: extract factors by principal components from $\mathbf{X}_t$ (Chapter 14); remove the part of the factors that moves with R_t within the period (Bernanke, Boivin and Eliasz split the series into "slow-moving" ones, which do not react within the month, and "fast-moving" ones such as asset prices); then estimate a VAR in $(\hat{\mathbf{F}}_t, R_t)$ and identify the policy shock recursively with R_t last [187].
2. **One-step Bayesian**: estimate factors and VAR jointly by Gibbs sampling, which propagates factor uncertainty into the impulse responses.

The response of any series in the panel follows from the loadings:

$$\frac{\partial \mathbf{X}_{t+h}}{\partial u_t} = \Lambda^{f} \frac{\partial \mathbf{F}_{t+h}}{\partial u_t} + \Lambda^{r} \frac{\partial R_{t+h}}{\partial u_t}.$$

28.3 THE PANEL VERSION

For a panel VAR across countries or regions, common factors extracted from the whole panel (global output, global interest rates, commodity prices) are added to each unit's equations. They control for cross-sectional dependence that would otherwise bias GMM estimates and confound domestic with global shocks [26].

28.4 APPLICATIONS

In Bernanke, Boivin and Eliasz's FAVAR on 120 US monthly series, a monetary tightening lowers industrial production and prices without the "price puzzle" (prices rising after a tightening) that small VARs often show, and the model delivers responses for consumption, employment, housing starts and many other series at once [187]. Panel versions are used to study global spillovers of US monetary policy and commodity shocks.

A SIMPLE EXAMPLE. A central bank watches 120 monthly series. A VAR in all 120 plus the policy rate would need $121 \times 121 = 14{,}641$ coefficients per lag. Summarising the 120 series by 3 principal-component factors gives a 4-variable VAR with 16 coefficients per lag, and the response of any of the 120 series is recovered from its factor loadings.

* * *

CHAPTER 29

PANEL REGIME SWITCHING

Panel threshold regression estimates regime-dependent coefficients with fixed effects:

$$y_{it} = \alpha_i + \beta_1' \mathbf{x}_{it} \mathbb{1}(q_{it} \le \gamma) + \beta_2' \mathbf{x}_{it} \mathbb{1}(q_{it} > \gamma) + \varepsilon_{it},$$

with the threshold chosen by least squares after the within transformation and tested by bootstrap [76]. Hansen's application asked whether financing constraints affect investment: with leverage as the threshold variable, the sensitivity of investment to cash flow differs between low- and high-debt firms. Markov-switching and smooth-transition panel versions follow the time-series models above [71, 169]; a well-known application is the effect of government debt on growth, with debt-to-GDP as the threshold variable.

29.1 DYNAMIC PANEL THRESHOLD MODELS

Hansen's panel threshold model assumes exogenous regressors and no lagged dependent variable. Seo and Shin allow both, estimating threshold models by first-difference GMM, with the threshold variable itself possibly endogenous [250]. Kremer, Bick and Nautz used a dynamic panel threshold model for 124 countries and found that inflation above about 2% in industrial and 17% in developing economies is associated with lower growth [251].

A SIMPLE EXAMPLE. An illustration: regress growth on government debt with a threshold at 60% of GDP. If the estimated slopes are -0.01 below the threshold and -0.05 above, a 10-percentage-point rise in debt costs 0.1 points of growth for a low-debt country but 0.5 points for a high-debt one. The panel threshold model estimates both the slopes and where the threshold lies, rather than assuming 60%.

> IN PLAIN WORDS. Panel threshold models ask whether a relationship changes once a unit crosses a line, such as a debt level, and estimate where that line is.

* * *

CHAPTER 30

TIME SERIES IN THE FREQUENCY DOMAIN

Instead of asking how today relates to yesterday, ask which *cycles* make up the series. The spectral density $f(\omega) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\gamma_k e^{-i\omega k}$ distributes the variance of a stationary series across frequencies $\omega \in [0,\pi]$, a cycle of period $2\pi/\omega$; the area under it equals the variance. The sample periodogram is $I(\omega_j) = \frac{1}{2\pi T}\left|\sum_{t=1}^{T} y_t e^{-i\omega_j t}\right|^2$ at the Fourier frequencies $\omega_j = 2\pi j/T$ [153]. **Analogy.** A prism splits white light into colours; the spectrum splits a time series into cycles.

30.1 SPECTRAL ANALYSIS

Decomposes variance by frequency and identifies cycles; computed by the fast Fourier transform [34, 153]. A white-noise series has a flat spectrum; an AR(1) with $\phi > 0$ has most power at low frequencies; seasonal series have peaks at the seasonal frequency and its harmonics.

R EXAMPLE. The smoothed periodogram of annual sunspot numbers peaks at a period of 11.1 years (the solar cycle), and that of the $\log_{10}$ lynx series at 10.0 years: the predator–prey cycle.

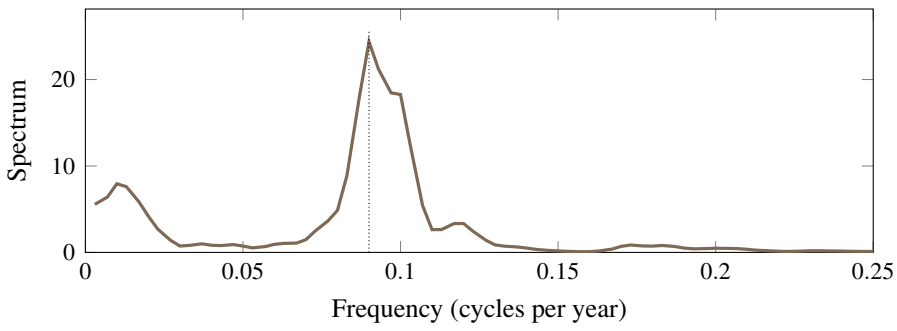


Figure 30.1: Smoothed periodogram of annual sunspot numbers (in thousands). The peak at frequency 0.09 cycles per year corresponds to a period of $1/0.09 \approx 11$ years, the solar cycle.

30.2 FILTERS

Separate trend from cycle. The Hodrick–Prescott filter chooses the trend τ_t to minimise $\sum (y_t - \tau_t)^2 + \lambda \sum (\Delta^2 \tau_t)^2$, with $\lambda = 1600$ for quarterly data [86]. Band-pass filters keep only cycles between chosen periods, typically 6 to 32 quarters for business cycles: Baxter–King (symmetric moving average) [11] and Christiano–Fitzgerald (asymmetric, usable at the end of the sample) [33]. Hamilton argues that the HP filter creates spurious cycles and distorts end-points, and proposes a simple regression alternative: the cycle is the error from regressing y_{t+8} on $y_t, \dots, y_{t-3}$ [73]. Central banks use such filters to estimate the output gap and the credit-to-GDP gap (the Basel III countercyclical buffer guide uses an HP filter with a very large λ).

30.3 POPULATION SPECTRUM AND PERIODOGRAM

The periodogram is an *inconsistent* estimator of the spectrum: its variance does not shrink as T grows, because each ordinate uses essentially two degrees of freedom. Smoothing across neighbouring frequencies, or weighting autocovariances, is needed [153].

30.4 LAG WINDOWS

Smooth the autocovariances: $\hat{f}(\omega) = \frac{1}{2\pi} \sum_{|k|<M} w(k/M) \hat{\gamma}_k e^{-i\omega k}$. A larger truncation point M gives less bias and more variance; the window shape trades leakage against resolution.

30.4.1 BARTLETT WINDOW

$w(u) = 1 - |u|$, a triangle [10]; it guarantees a non-negative estimate and is the weighting used in Newey–West standard errors.

30.4.2 TUKEY WINDOW

Tukey's family of cosine windows, $w(u) = 1 - 2a + 2a\cos \pi u$ [15].

30.4.3 DANIELL WINDOW

A rectangular (moving-average) smoother of the periodogram in the frequency domain, whose lag window is $w(u) = \sin(\pi u)/(\pi u)$ [153].

30.4.4 TUKEY–HANNING WINDOW

$w(u) = \frac{1}{2}(1+\cos \pi u)$, the member $a = 1/4$ of the Tukey family [15].

A SIMPLE EXAMPLE. A monthly series with a strong 12-month seasonal cycle has a spectral peak at frequency $\omega = 2\pi/12 = 0.52$ radians per month (and harmonics at 2ω, 3ω). A quarterly GDP cycle of 8 years (32 quarters) peaks at $\omega = 2\pi/32 = 0.20$. Reading the period off a spectrum is just $2\pi/\omega$.

> IN PLAIN WORDS. The frequency domain asks "how much of the series' movement comes from cycles of each length?" and answers with a spectrum, much as a sound engineer reads bass and treble.

* * *

Chapter 31

Cross-Spectral Analysis

Measures the relation between two series frequency by frequency, using the cross spectrum $f_{xy}(\omega)$. Three quantities [63, 153]:

- **Coherence** $\kappa^2(\omega) = |f_{xy}(\omega)|^2 / \{f_{xx}(\omega) f_{yy}(\omega)\}$, an R^2 by frequency: two series can be closely related at business-cycle frequencies and unrelated at high frequencies;
- **Gain**: by how much y's cycle of a given length is amplified relative to x's;
- **Phase**: the lead or lag at each frequency (phase shift divided by ω gives the time lag).

Application. Coherence between money growth and inflation is high at low frequencies (the long run) and weak at high frequencies, a frequency-domain statement of the quantity theory.

31.1 Short-time Fourier transform and Gabor transform

Ordinary spectra assume the cycles are stable over the whole sample. The windowed Fourier transform $V(\tau, \omega) = \int y(t) w(t - \tau) e^{-i\omega t}\, dt$ computes a spectrum in a moving window centred at τ, giving a time–frequency picture; the Gabor transform uses a Gaussian window [53]. Its limitation is a fixed window length: the same time resolution for short and long cycles, which motivates wavelets.

A simple example. Money growth and inflation might have coherence 0.8 at an 8-year cycle but only 0.1 at a 1-year cycle: they move together in the long run, not month to month. If the phase at the 8-year frequency corresponds to a lag of 2 quarters, money leads inflation by about half a year at that horizon.

* * *

CHAPTER 32

WAVELET TYPES

Wavelets localise in both time and frequency: short windows for short cycles, long windows for long ones. The continuous transform is $W(a,b) = \int y(t)\, a^{-1/2} \psi^*\left(\frac{t-b}{a}\right) dt$, with scale a (cycle length) and location b (time). The discrete transform expands the series in a father wavelet ϕ (smooth part) and mother wavelets ψ (details at each scale j): $y_t = S_J + D_J + \cdots + D_1$, a multiresolution analysis [35, 36, 121, 142]. **Analogy.** A musical score: it tells you which notes (frequencies) are played and *when*, which a spectrum cannot.

Applications. Decomposing returns into investment horizons (a daily trader and a pension fund care about different scales); estimating betas and correlations scale by scale; tracking how the effect of monetary policy on output changes over time and across frequencies [205]; and wavelet coherence maps of co-movement between markets that show contagion appearing in crises at short scales [67, 175].

32.1 DISCRETE WAVELETS

Used for multiresolution analysis, denoising and scale-by-scale regressions [56]. The maximal-overlap DWT (MODWT) is preferred in economics because it works for any sample size and is shift-invariant [142].

32.1.1 BEYLKIN

Filters designed by Beylkin, Coifman and Rokhlin for fast numerical computation: applying an operator (such as a differential or integral operator) in a wavelet basis produces a sparse matrix, which speeds up computation [13]. In economics they are mainly an option in software rather than a tool chosen for interpretation.

32.1.2 BNC

As labelled in the outline (a software-dependent name, likely a battle–Lemarié or B-spline family in a particular package); see [36].

32.1.3 COIFLET

Daubechies' wavelets designed at Coifman's request, with vanishing moments for both the wavelet ψ and the scaling function ϕ; they are nearly symmetric, which reduces phase distortion when the series is reconstructed from its components [36].

32.1.4 HAAR

The simplest wavelet, a step function: averages and differences of adjacent pairs [68]. Its detail coefficient at scale 1 is $(y_{2t} - y_{2t-1})/\sqrt{2}$, a scaled first difference.

32.1.5 SYMLET

Nearly symmetric variants of Daubechies wavelets, with the same number of vanishing moments [36]; the least-asymmetric LA(8) is a common default.

32.2 CONTINUOUS WAVELETS

The continuous transform evaluates $W(a,b)$ on a fine grid of scales and locations, giving a smooth time–scale map (a scalogram) rather than a few discrete levels; it is redundant, which makes it good for visual analysis and coherence, but not for exact reconstruction or regressions [122].

32.3 WAVELET FUNCTIONS

Complex-valued functions suited to time–frequency analysis, because their modulus gives power and their argument gives phase; used for wavelet coherence and phase-difference (lead–lag) analysis [67, 175].

32.3.1 COMPLEX MEXICAN HAT

The Mexican hat is the negative second derivative of a Gaussian, a symmetric "sombrero"; its complex extension adds an imaginary part (via the Hilbert transform) so that phase can be measured [122]. Good time localisation, broad frequency band.

32.3.2 FBSP (FREQUENCY B-SPLINE)

A spline wavelet whose bandwidth, centre frequency and order can be chosen separately, so the user can tune the trade-off between time and frequency resolution [122].

32.3.3 MORLET

A Gaussian-modulated complex exponential, $\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2}$ with $\omega_0 = 6$ the usual choice; it balances time and frequency resolution well and is the most widely used wavelet in economics and geophysics [126, 175].

32.3.4 SHANNON

The ideal band-pass wavelet derived from sampling theory, $\psi(t) = \operatorname{sinc}(t/2)\cos(3\pi t/2)$: it keeps exactly one frequency band, so it has perfect frequency localisation but decays slowly in time [161]. A benchmark rather than a working choice for economic data.

32.4 MULTIVARIATE WAVELET TOOLS

Partial and multiple wavelet coherence extend coherence to three or more series, e.g. the co-movement of two stock markets after controlling for oil prices [252].

A SIMPLE EXAMPLE. Haar transform of four observations $4, 6, 10, 12$. Level 1: smooth coefficients $(4+6)/\sqrt{2} = 7.07$ and $(10+12)/\sqrt{2} = 15.56$; details $(6-4)/\sqrt{2} = 1.41$ and $(12-10)/\sqrt{2} = 1.41$. Level 2: smooth $(7.07+15.56)/\sqrt{2} = 16.0$, detail $(15.56-7.07)/\sqrt{2} = 6.0$. The "energy" is preserved: $4^2+6^2+10^2+12^2 = 296 = 16^2+6^2+1.41^2+1.41^2$. The large level-2 detail says most variation is a slow upward movement; the small level-1 details say the short-term wiggles are minor.

> IN PLAIN WORDS. Wavelets are a zoom lens for time series: they show which cycles were active, and when.

CHAPTER 33

BAYESIAN ECONOMETRICS

Posterior $\propto$ likelihood $\times$ prior: $p(\theta \mid \mathbf{y}) \propto p(\mathbf{y} \mid \theta)\, p(\theta)$ [101]. The prior encodes what is known before the data; the posterior updates it. Uncertainty is stated directly as probability ("there is a 95% probability that β lies in this interval"), and prior information regularises models with many parameters.

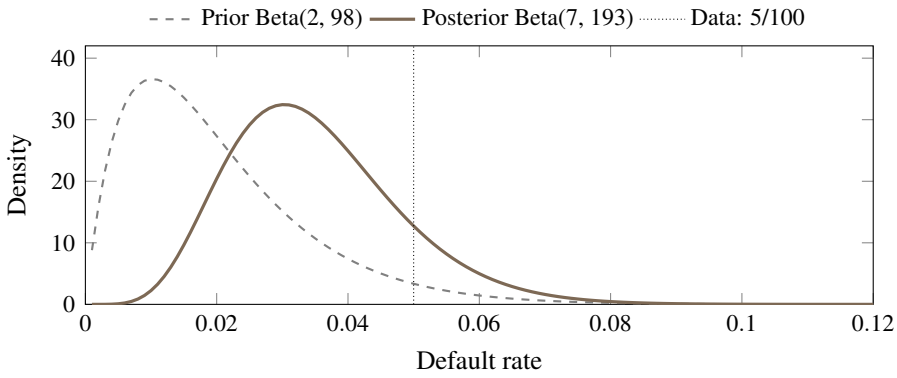


Figure 33.1: Bayesian updating of a default rate (simple example of this chapter): the posterior (solid) moves from the prior mean of 2% towards the observed 5%, ending at 3.5%, and is narrower than the prior.

33.1 BAYESIAN VECTOR AUTOREGRESSION (BVAR)

Shrinks VAR coefficients towards a simple benchmark, the *Minnesota prior*: each variable is a random walk a priori (own first lag 1, other coefficients 0), with prior variances that shrink more for longer lags and for other variables' lags [39, 111]. This allows large systems without over-fitting; conjugate and hierarchical versions choose the tightness from the data [58, 96, 165]. Litterman's BVAR forecasts at the Federal Reserve Bank of Minneapolis were competitive with large commercial macroeconomic models [111], and BVARs remain standard forecasting tools at central banks.

33.2 PRIORS AND POSTERIOR SIMULATION

When posteriors are not available analytically they are simulated by Markov chain Monte Carlo: construct a Markov chain whose stationary distribution is the posterior, run it, discard the burn-in, and use the draws [30].

33.2.1 METROPOLIS SAMPLING

Propose $\theta^* = \theta +$ symmetric noise; accept with probability $\min\{1, p(\theta^* \mid \mathbf{y})/p(\theta \mid \mathbf{y})\}$, otherwise stay [125]. Moves uphill are always accepted, downhill sometimes, so the chain explores the posterior in proportion to its height.

33.2.2 METROPOLIS–HASTINGS ALGORITHM

Generalises Metropolis to asymmetric proposal densities q, with acceptance probability [79]

$$\min\left\{1,\ \frac{p(\theta^* \mid \mathbf{y})\, q(\theta \mid \theta^*)}{p(\theta \mid \mathbf{y})\, q(\theta^* \mid \theta)}\right\}.$$

R EXAMPLE. A random-walk Metropolis sampler for the regression of car stopping distance on speed (R's `cars` data, flat priors, 20,000 draws, 5,000 burn-in) accepts 43% of proposals and gives a posterior mean slope of 3.96 with a 95% credible interval $[3.21, 4.76]$, close to the OLS estimate of 3.93, as it should be with flat priors.

33.2.3 MARKOV CHAIN MONTE CARLO (MCMC)

Gibbs sampling is the special case that draws each block of parameters from its full conditional distribution, with acceptance probability one [54, 55]. It underlies Bayesian estimation of factor models, Markov-switching models and stochastic volatility. Convergence should be checked with trace plots, several chains and effective sample sizes.

33.2.4 SLICE SAMPLING

An auxiliary-variable sampler: draw a height under the density, then sample uniformly from the "slice" of parameter values above that height; it adapts its step size and needs no tuning of a proposal density [130].

33.3 LARGE BVARS AND TIME-VARYING PARAMETERS

Bańbura, Giannone and Reichlin showed that a BVAR with 130 variables, with shrinkage tightened as the system grows, forecasts better than small VARs and factor models [253]. Primiceri's time-varying-parameter VAR with stochastic volatility lets coefficients and shock variances drift, and was used to ask whether US monetary policy or smaller shocks explained the Great Moderation [254].

33.4 HAMILTONIAN MONTE CARLO

Random-walk Metropolis struggles in high dimensions. Hamiltonian Monte Carlo uses the gradient of the log posterior to propose distant moves that are still likely to be accepted; the No-U-Turn Sampler tunes the path length automatically [255], and is the default sampler in Stan.

A SIMPLE EXAMPLE. A bank believes a new loan segment's default rate is about 2%, worth the evidence of 100 loans: prior $\text{Beta}(2,98)$, mean $2/100$. In the first 100 loans, 5 default. The posterior is $\text{Beta}(2+5,98+95) = \text{Beta}(7,193)$, with mean $7/200 = 3.5\%$, between the prior (2%) and the data (5%). With 1,000 loans and 50 defaults the posterior mean would be $52/1{,}100 = 4.7\%$: as data accumulate, the prior matters less.

> IN PLAIN WORDS. Bayesian econometrics starts from what you believed, lets the data change your mind, and reports the result as a probability distribution rather than a single number.

* * *

CHAPTER 34

WHICH METHOD WHEN

Question	Start with
Does x affect y in a cross-section?	OLS with diagnostics (Ch. 1); IV if endogenous
Is the series stationary?	ADF + KPSS (Ch. 3); break tests if a known event looms (Ch. 4)
More than two breaks?	Count with Bai–Perron / Kejriwal–Perron, then Carrion-i-Silvestre–Kim–Perron (4.7–4.11); Maki or Kejriwal–Perron for cointegration (6.10–6.11); Fourier tests for smooth change
Is there a bubble?	GSADF test of Phillips, Shi and Yu (3.8)
Do $I(1)$ series share a long run?	Engle–Granger, Johansen, ARDL bounds (Ch. 5); Gregory–Hansen if shifts (Ch. 6)
Does x help forecast y?	Granger or Toda–Yamamoto (Ch. 7); Dumitrescu–Hurlin in panels
How do shocks propagate in a system?	VAR/VECM with IRFs or local projections (Ch. 8–9); SVAR, proxy SVAR for named shocks (Ch. 10)
Hundreds of predictors?	Factor models, PCA (Ch. 13–14); BVAR (Ch. 33)
Do dynamics change with the state of the economy?	Linearity tests (Ch. 19), then TAR, STAR or Markov switching (Ch. 16–18)
Does risk change over time?	ARCH test, GARCH family, DCC; VaR backtests (Ch. 20)
Many units over time?	FE/RE with Hausman, dynamic GMM, PMG, CCE (Ch. 21); CIPS and panel cointegration tests (Ch. 22–24)
Did a policy introduced at different dates work?	Staggered difference-in-differences (21.9)
Which cycles drive the data, and when?	Spectrum, filters, cross-spectra (Ch. 30–31); wavelets (Ch. 32)

Table 34.1: A quick map of the outline.

* * *

CHAPTER 35

CORRECTIONS TO THE ORIGINAL OUTLINE

Carried over from the October 2026 version. Several names and items in the original outline were corrected or flagged: Tsey → Tsay; Philips → Phillips; Elliot → Elliott; Luptaphol → Lütkepohl; Saikkonam → Saikkonen; Turkey → Tukey; Barlett → Bartlett; Coifler → Coiflet; Slides sampling → slice sampling; Weterlund → Westerlund; Levin and Lee → Levin and Lin; Haris and Tzavalis → Harris and Tzavalis; Smoothing models → smooth transition models. Item 3.5 repeated "ARIMA"; the Ng–Perron tests were used instead. Items marked "as in the outline" could not be matched to a specific source.

Added in this revision.

- **20.1.6 VaR.** The October version wrote $\mathrm{VaR}_t(\alpha) = \mu_t + \sigma_t q_\alpha$, which is the return quantile (a negative number for small α). Stated as a loss, $\mathrm{VaR}_t(\alpha) = -(\mu_t + \sigma_t q_\alpha)$.
- **19.1.1 RESET.** Its time-series descendants include Tsay's (1986) F test, distinct from the Tsay (1989) arranged-regression threshold test of item 19.1.2; reference 202 was added.
- **10. SVAR.** With $\mathbf{A}\varepsilon_t = \mathbf{B}\mathbf{u}_t$ and $\mathbf{A} = \mathbf{I}$, $K(K-1)/2$ restrictions on $\mathbf{B}$ are needed; in the general AB-model the number is larger, because both matrices must be restricted.
- **30.4.3 Daniell window.** The Daniell window is a rectangular smoother *in the frequency domain*; $\sin(\pi u)/(\pi u)$ is its equivalent lag window.
- **6.8 Narayan and Popp** and **32.1.2 BNC** remain unconfirmed.
- **New items.** 39 items and references 206–259 were added to bring the outline up to date; see "How to Use This Outline". The references for the new items were checked against publisher or RePEc records where possible.
- The DOIs in references 1–190 were carried over from the October version, which notes

that they were entered from memory; spot-check them before circulating.

* * *

APPENDIX: R CODE FOR THE EXAMPLES

Every R EXAMPLE in the text is produced by this script. It uses only data sets that ship with R and no add-on packages.

```
# ====================================================================
# Outline of Econometric Models: worked examples on real data
# Every example uses a data set that ships with R; no add-on packages.
# ====================================================================

set.seed(2018)
adf_t <- function(y, k = 1, trend = FALSE, dummies = NULL) {
  # ADF regression: dy_t = a (+ b t) + d y_{t-1} + sum psi_j dy_{t-j} (+ dummies)
  n <- length(y); dy <- diff(y)
  Y <- dy[(k + 1):(n - 1)]; ylag <- y[(k + 1):(n - 1)]
  L <- if (k > 0) sapply(1:k, function(j) dy[(k + 1 - j):(n - 1 - j)]) else NULL
  X <- cbind(ylag, L)
  if (trend) X <- cbind(X, tt = (k + 2):n)
  if (!is.null(dummies)) X <- cbind(X, dummies[(k + 2):n, , drop = FALSE])
  summary(lm(Y ~ X))$coefficients[2, 3]
}

# ---- 1. Classical regression and its violations: Longley data -----
# US macro data 1947-62, a famous multicollinearity test case
m <- lm(Employed ~ ., data = longley)
print(round(summary(m)$coefficients, 4)); cat("R2 =", round(summary(m)$r.squared, 4),
    "\n")
X <- longley[, names(longley) != "Employed"]
vif <- sapply(names(X), function(v) 1 / (1 - summary(lm(X[[v]] ~ ., data = X[names(X)
    != v]))$r.squared))
print(round(vif, 1))
e <- resid(m); n <- length(e)
cat("Durbin-Watson =", round(sum(diff(e)^2) / sum(e^2), 3), "\n")
bp <- n * summary(lm(I(e^2) ~ ., data = X))$r.squared             # Breusch-Pagan LM
    (Koenker)
cat("Breusch-Pagan LM =", round(bp, 2), " p =", round(pchisq(bp, ncol(X), lower.tail
```

```
    = FALSE), 3), "\n")
S <- mean(e^3) / mean(e^2)^1.5; K <- mean(e^4) / mean(e^2)^2
jb <- n / 6 * (S^2 + (K - 3)^2 / 4)
cat("Jarque-Bera =", round(jb, 3), " p =", round(pchisq(jb, 2, lower.tail = FALSE),
    3), "\n")

# ---- 2. Dummy variables: the UK seat-belt law of 31 January 1983 ---
sb <- data.frame(deaths = as.numeric(Seatbelts[, "DriversKilled"]),
                 petrol = as.numeric(Seatbelts[, "PetrolPrice"]),
                 km     = as.numeric(Seatbelts[, "kms"]),
                 law    = as.numeric(Seatbelts[, "law"]),
                 month  = factor(cycle(Seatbelts)), t = seq_len(nrow(Seatbelts)))
ms <- lm(log(deaths) ~ law + log(petrol) + log(km) + month + t, data = sb)
b_law <- coef(ms)["law"]
cat("\nSeat-belt law coefficient =", round(b_law, 4), " s.e. =",
    round(summary(ms)$coefficients["law", 2], 4),
    "  => change in deaths =", round(100 * (exp(b_law) - 1), 1), "%\n")

# ---- 3. Box-Jenkins: the airline model ----------------------------
air <- log(AirPassengers)
fit <- arima(air, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1), period =
    12))
print(fit)
cat("Ljung-Box (24 lags, 2 params) p =",
    round(Box.test(resid(fit), 24, "Ljung-Box", fitdf = 2)$p.value, 3), "\n")
fc <- predict(fit, 12)
cat("Forecast passengers (000s), Jan-Dec 1961:\n"); print(round(exp(fc$pred)))

# ---- 4. Unit roots and a structural break: the Nile ----------------
nile <- as.numeric(Nile)                        # annual flow at Aswan, 1871-1970
cat("\nADF t (constant, 1 lag) =", round(adf_t(nile, 1), 3), " (5% cv -2.89)\n")
# Zivot-Andrews (model A, level shift): minimise ADF t over break dates
yrs <- 1871:1970; cand <- 16:85
za <- sapply(cand, function(b) { DU <- matrix(as.numeric(seq_along(nile) > b), ncol =
     1)
  adf_t(nile, 1, dummies = DU) })
cat("Zivot-Andrews min t =", round(min(za), 3), " at break after", yrs[cand[which.min
    (za)]],
    " (5% cv -4.80)\n")
cat("Mean flow before / after 1898:", round(mean(nile[yrs <= 1898])), "/",
    round(mean(nile[yrs > 1898])), "\n")

# ---- 5. Cointegration and causality: European stock indices --------
ls <- log(EuStockMarkets)
eg <- lm(ls[, "DAX"] ~ ls[, "CAC"])
cat("\nEngle-Granger: DAX on CAC slope =", round(coef(eg)[2], 3),
```

```
    "  ADF t on residuals =", round(adf_t(resid(eg), 1), 3), " (5% cv -3.34)\n")
r <- 100 * diff(ls)
granger <- function(y, x, p = 2) {
  n <- length(y); Y <- y[(p + 1):n]
  Ly <- sapply(1:p, function(j) y[(p + 1 - j):(n - j)])
  Lx <- sapply(1:p, function(j) x[(p + 1 - j):(n - j)])
  a <- anova(lm(Y ~ Ly), lm(Y ~ Ly + Lx)); c(F = a$F[2], p = a$`Pr(>F)`[2])
}
cat("Granger: FTSE -> DAX"); print(round(granger(r[, "DAX"], r[, "FTSE"]), 4))
cat("Granger: DAX -> FTSE"); print(round(granger(r[, "FTSE"], r[, "DAX"]), 4))

# ---- 6. Factors: principal components of four stock markets --------
pc <- prcomp(r, scale. = TRUE)
cat("\nShare of variance by component:", round(summary(pc)$importance[2, ], 3), "\n")
cat("Loadings on PC1:", round(pc$rotation[, 1], 3), "\n")

# ---- 7. Threshold autoregression: the lynx cycle -------------------
y <- log10(as.numeric(lynx)); n <- length(y)
Y <- y[3:n]; y1 <- y[2:(n - 1)]; y2 <- y[1:(n - 2)]
ssr_thr <- function(r) { lo <- y2 <= r
  sum(resid(lm(Y ~ (y1 + y2) * lo))^2) }
grid <- sort(y2)[round(0.15 * length(y2)):round(0.85 * length(y2))]
ssr <- sapply(grid, ssr_thr); r_hat <- grid[which.min(ssr)]
ssr_lin <- sum(resid(lm(Y ~ y1 + y2))^2)
cat("\nSETAR(2;2,2), delay 2: threshold =", round(r_hat, 3),
    "  SSR linear =", round(ssr_lin, 3), "  SSR threshold =", round(min(ssr), 3), "\n
    ")
lo <- y2 <= r_hat
print(round(rbind(lower = coef(lm(Y[lo] ~ y1[lo] + y2[lo])),
                  upper = coef(lm(Y[!lo] ~ y1[!lo] + y2[!lo]))), 3))

# ---- 8. Frequency domain: sunspot and lynx cycles ------------------
peak <- function(x) { s <- spec.pgram(x, spans = c(3, 3), plot = FALSE, detrend =
    TRUE)
  1 / s$freq[which.max(s$spec)] }
cat("\nPeak period: sunspots =", round(peak(sunspot.year), 1), "years;  lynx =",
    round(peak(log10(lynx)), 1), "years\n")

# ---- 9. Panel data: fixed vs random effects (simulated firms) -------
N <- 200; T <- 6
a  <- rnorm(N)                                  # firm effect
x  <- matrix(0.8 * a + rnorm(N * T), N, T)        # regressor correlated with a
yv <- 1 + 0.5 * x + a + matrix(rnorm(N * T), N, T)
df <- data.frame(id = rep(1:N, T), y = as.vector(yv), x = as.vector(x))
pool <- coef(lm(y ~ x, df))[2]
dm <- function(v) v - ave(v, df$id)
```

```
fe <- lm(dm(df$y) ~ dm(df$x) - 1); b_fe <- coef(fe)
# Random effects by quasi-demeaning (Swamy-Arora variance components)
s2e <- sum(resid(fe)^2) / (N * T - N - 1)
mb  <- lm(ave(y, id) ~ ave(x, id), df[!duplicated(df$id), ])
s2b <- sum(resid(mb)^2) / (N - 2); s2a <- max(s2b - s2e / T, 0)
th  <- 1 - sqrt(s2e / (s2e + T * s2a))
qd  <- function(v) v - th * ave(v, df$id)
re  <- lm(qd(df$y) ~ I(rep(1 - th, N * T)) + qd(df$x) - 1)
b_re <- coef(re)[2]
v_fe <- vcov(fe)[1, 1] * (N * T - 1) / (N * T - N - 1); v_re <- vcov(re)[2, 2]
H <- (b_fe - b_re)^2 / (v_fe - v_re)
cat("\nTrue slope 0.5: pooled OLS =", round(pool, 3), " FE =", round(b_fe, 3),
    " RE =", round(b_re, 3), "  Hausman =", round(H, 1), " p =",
    signif(pchisq(H, 1, lower.tail = FALSE), 3), "\n")

# ---- 10. Bayesian estimation by Metropolis-Hastings ----------------
# Regression of stopping distance on speed (cars data), flat priors
yc <- cars$dist; xc <- cars$speed
lpost <- function(th) { if (th[3] <= 0) return(-Inf)
  sum(dnorm(yc, th[1] + th[2] * xc, th[3], log = TRUE)) - log(th[3]) }
draws <- matrix(NA, 20000, 3); th <- c(-17, 4, 15); lp <- lpost(th); acc <- 0
for (i in 1:20000) {
  prop <- th + rnorm(3, 0, c(3, 0.2, 1))
  lpp <- lpost(prop)
  if (log(runif(1)) < lpp - lp) { th <- prop; lp <- lpp; acc <- acc + 1 }
  draws[i, ] <- th
}
keep <- draws[-(1:5000), ]
cat("\nMH acceptance rate =", round(acc / 20000, 2), "\n")
print(round(rbind(OLS = c(coef(lm(dist ~ speed, cars)), summary(lm(dist ~ speed, cars
    ))$sigma),
                  posterior_mean = colMeans(keep),
                  lower_95 = apply(keep, 2, quantile, 0.025),
                  upper_95 = apply(keep, 2, quantile, 0.975)), 3))
```

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